INVOLUTIVE PROOFS OF VARIOUS COMBINATORIAL IDENTITIES FOR EULER NUMBERS

SHISHUO FU, YANG LI, ZHICONG LIN, AND YONGCHUN ZANG

Abstract. We provide involutive proofs of various combinatorial identities for Euler numbers, including Carlitz's symmetric function formulas for up-down permutations of a multiset and identities involving Stirling numbers, powers of 2 or 4 and Legendre transformation. In the course, q-analogs of known or new identities for Euler numbers are derived from the combinatorial perspective.

1. Introduction

The *Euler numbers*, denoted E_n, can be defined as the coefficients of the Taylor expansion:

$$\tan(x)+\sec(x)=\sum_{n\geqslant 0}E_n\frac{x^n}{n!}=1+x+1\frac{x^2}{2!}+2\frac{x^3}{3!}+5\frac{x^4}{4!}+16\frac{x^5}{5!}+61\frac{x^6}{6!}+\cdots.$$

Euler numbers have been extensively studied in Combinatorics and Number Theory; see Stanley's survey [34], Stanley's talk on the 14th Ramanujan Colloquium [35] and Sun's talk on Bernoulli Numbers and Euler Numbers [38]. The numbers E_n with even indices and odd indices are also known as *secant numbers* and *tangent numbers*, respectively.

Motivated by recent works of Sagan [29], Zhao, Lin and Zang [40], and Kim and Lin [21] on combinatorial proofs of identities involving Euler numbers, we streamline a combinatorial investigation of the following identities and their companions:

$$\sum_{k=0}^{n-1}\binom{2n}{2k+1}2^{2n-2k}(-1)^k E_{2k+1}=2^{2n+1}, \tag{1.1}$$

$$\sum_{k=1}^{2n+1}2^{2n+1-k}(-1)^k k!S(2n+1,k)=(-1)^{n+1}E_{2n+1}, \tag{1.2}$$

$$\sum_{k=0}^{n}\binom{2n+1}{2k+1}(-1)^k 2^{2n-2k}E_{2n-2k}=S_{2n+1}, \tag{1.3}$$

$$\sum_{k=0}^{n}\binom{4n}{4k}(-4)^k E_{4n-4k}=1, \tag{1.4}$$

$$\sum_{k=0}^{n}\binom{4n+1}{4k}(-4)^{k-n}E_{4n-4k+1}=1, \tag{1.5}$$

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$$\sum_{k=0}^{n}\binom{2n}{2k}\binom{2n+2k}{2k}2^{2n-2k}(-1)^k E_{2k}=(-1)^n\binom{2n}{n}, \tag{1.6}$$

where $S(n,k)$ and S_n denote the *Stirling numbers of the second kind* and the *Springer numbers*, respectively. Note that (1.1) was originally conjectured by Aliniaeifard and Li [1] and later proved combinatorially by Zhao, Lin and Zang [40] along with its q-analog; (1.2) is a consequence of the famous Frobenius formula [14] and Euler's signed formula [11]; (1.3) was proved analytically by Han et al. [17]; (1.4) was derived by Hirschhorn [18] using generating functions, while its tangent-number companion (1.5) is new; (1.6) was established algebraically by Sun in [37]. Using a sign-reversing involution is a common way to give a combinatorial proof of identities (just like (1.1)–(1.6)) involving a summation with alternating sign. Such an approach was summarized nicely as "Description, Involution, Exception" by Benjamin and Quinn in [5]. Our goal here is to construct an appropriate "killing involution" for each of the identities mentioned above. The main contributions of this paper can be outlined as follows.

- By extending the involution proposed by Sagan [29], we present in Sec. 2 a combinatorial proof of Carlitz's symmetric function formulas [6] for counting up-down permutations of a multiset.
- In Sec. 3, we construct Sagan-style involutions for q-analogs of (1.2) and its secant-number companion (see (3.4)). As applications, we provide alternative combinatorial proofs of q-analogs of (1.1) and three of its companions.
- Based on the combinatorial model of snakes, involutive proofs of generalizations of (1.3) and its tangent-number companion are provided in Sec. 4.
- Sec. 5 is devoted to the combinatorics of Hirschhorn-like identities. In particular, two distinct q-analogs of (1.5) are found, along with their respective interesting involutive proofs.
- Sec. 6 is devoted to the combinatorics of Sun-like identities. We split our combinatorial proof of Sun's secant-number identity (1.6) into three steps, and this approach yields three additional Sun-like identities. However, the problem of deriving a q-analog of (1.6) remains open. In the course, a new alternating sum formula for Catalan numbers is found (see (6.32)), which may be of independent interest.

Finally, we conclude our paper with some related open problems.

2. On Carlitz's symmetric function formulas for up-down permutations of a multiset

Let $\lambda=(\lambda_1,\lambda_2,\ldots,\lambda_n)$ be a composition with $|\lambda|=\lambda_1+\lambda_2+\cdots+\lambda_n$. Let $\mathfrak{S}_\lambda$ denote the set of all permutations of the multiset $\{1^{\lambda_1},2^{\lambda_2},\ldots,n^{\lambda_n}\}$.

Definition 2.1. *A (multi-)permutation $\pi=\pi_1\cdots\pi_{|\lambda|}\in\mathfrak{S}_\lambda$ is up-down (or alternating) if*

$$\pi_1\leqslant\pi_2>\pi_3\leqslant\pi_4>\pi_5\leqslant\cdots.$$

If the above inequalities are reversed, then π is referred to as a down-up permutation. Denote by $\mathcal{A}_\lambda$ the set of all up-down permutations in $\mathfrak{S}_\lambda$. For simplicity, we write $\mathfrak{S}_\lambda=\mathfrak{S}_n$ and $\mathcal{A}_\lambda=\mathcal{A}_n$ when $\lambda=(1^n):=(\underbrace{1,\ldots,1}_{n\ copies})$.

For example, we have

$$\mathcal{A}_{(2^2)} = \{1212, 2211\} \quad \text{and} \quad \mathcal{A}_{(1,2^2)} = \{13232, 33221, 23132, 23231\}.$$

Recall that the *complete symmetric function* h_n is defined by $h_0 := 1$, and for $n \geqslant 1$,

$$h_n = \sum_{i_1 \leqslant i_2 \leqslant \cdots \leqslant i_n} x_{i_1} x_{i_2} \cdots x_{i_n}.$$

Carlitz [7] proved the following symmetric function formulas for counting up-down multipermutations.

Theorem 2.2 (Carlitz). *Fix $\lambda = (\lambda_1, \ldots, \lambda_n)$.*

(1) If $|\lambda|$ is even, then $|\mathcal{A}_\lambda|$ equals the coefficient of $x_1^{\lambda_1} x_2^{\lambda_2} \cdots x_n^{\lambda_n}$ in

$$E := \frac{1}{\sum_{k \geqslant 0} (-1)^k h_{2k}}.$$

(2) If $|\lambda|$ is odd, then $|\mathcal{A}_\lambda|$ equals the coefficient of $x_1^{\lambda_1} x_2^{\lambda_2} \cdots x_n^{\lambda_n}$ in

$$T := \frac{\sum_{k \geqslant 0} (-1)^k h_{2k+1}}{\sum_{k \geqslant 0} (-1)^k h_{2k}}.$$

Applying the *exponential specialization* ex (see [32, Sec. 7.8]) to E and T with $\lambda = (1^n)$ gives the exponential generating function for up-down permutations

$$\sum_{n \geqslant 0} |\mathcal{A}_n| \frac{t^n}{n!} = \sec(t) + \tan(t),$$

a result attributed to André [2]. Moreover, Carlitz's formulas were applied by Gessel [15] and Stanely [33] to enumerate some special up-down multipermutations. In particular, Gessel [15] proved via some scalar product formulas that

$$|\mathcal{A}_{(1^m, 2^n)}| = \begin{cases} 2^{-n} \sum_{k=0}^{n} \binom{n}{k} E_{2k+m}, & \text{if } m \text{ is even;} \\ 2^{-n} E_{2n+m}, & \text{if } m \text{ is odd.} \end{cases}$$

Inspired by a recent work of Sagan [29], we aim to provide an involutive proof of Theorem 2.2.

Involution proof of Theorem 2.2. Let $M = \{1^{\lambda_1}, 2^{\lambda_2}, \ldots, n^{\lambda_n}\}$. We begin with the proof of the case for $|\lambda|$ being even.

An ordered (multi-)set partition $\pi = (B_1, B_2, \ldots, B_k)$ of M with each block B_i having even size is written as $\pi \vDash_2 M$ with the *number of blocks* denoted as $\mathrm{bk}(\pi) = k$. Since

$$E = 1 + \sum_{k \geqslant 1} (-1)^{k-1} h_{2k} + \left(\sum_{k \geqslant 1} (-1)^{k-1} h_{2k} \right)^2 + \left(\sum_{k \geqslant 1} (-1)^{k-1} h_{2k} \right)^3 + \cdots,$$

we have

$$[x^\lambda] E = \sum_{\pi \vDash_2 M} (-1)^{|\lambda|/2 - \mathrm{bk}(\pi)}, \tag{2.1}$$

where $x^\lambda := x_1^{\lambda_1}x_2^{\lambda_2}\cdots x_n^{\lambda_n}$. It remains to construct an involution ι on $\{\pi : \pi \vDash_2 M\}$ to show that

$$\sum_{\pi\vDash_2 M}(-1)^{|\lambda|/2-\mathrm{bk}(\pi)} = |\mathcal{A}_\lambda|. \tag{2.2}$$

A block B_i of $\pi = (B_1, B_2, \ldots, B_k) \vDash_2 M$ is said to be *splittable* if $|B_i| > 2$; and is *mergeable* if $|B_i| = 2$ and $\max(B_i) \leqslant \min(B_{i+1})$, where we make the convention that $\min(B_{k+1}) = 0$. Now define the involution ι on $\{\pi : \pi \vDash_2 M\}$ as:

- If π has neither splittable nor mergeable blocks, then $\iota(\pi) = \pi$;
- otherwise, let i be the smallest index such that the block $B_i = \{b_1 \leqslant b_2 \leqslant \cdots\}$ is either splittable or mergeable and define

$$\iota(\pi) = \begin{cases} (B_1, \ldots, B_{i-1}, \{b_1, b_2\}, B_i\backslash\{b_1, b_2\}, B_{i+1}, \ldots, B_k), & \text{if } B_i \text{ is splittable;} \\ (B_1, \ldots, B_{i-1}, B_i \cup B_{i+1}, B_{i+2} \ldots, B_k), & \text{if } B_i \text{ is mergeable.} \end{cases}$$

It is plain to see that ι is an involution on $\{\pi : \pi \vDash_2 M\}$ fixing elements in

$$\mathcal{B}_\lambda = \{\pi \vDash_2 M : \pi \text{ has neither splittable nor mergeable blocks}\}$$

and sign-reversing on $\{\pi : \pi \vDash_2 M\}\backslash\mathcal{B}_\lambda$. Thus,

$$\sum_{\pi\vDash_2 M}(-1)^{|\lambda|/2-\mathrm{bk}(\pi)} = \sum_{\pi\in\mathcal{B}_\lambda}(-1)^{|\lambda|/2-\mathrm{bk}(\pi)} = |\mathcal{A}_\lambda|,$$

where the last equality follows from the observation that a partition $\pi = (B_1, \ldots, B_k)$ lies in $\mathcal{B}_\lambda$ if and only if each block B_i has size 2 (i.e., $\mathrm{bk}(\pi) = |\lambda|/2$) and $\max(B_i) > \min(B_{i+1})$ for all $1 \leqslant i < k$. This proves (2.2) and thus completes the proof of the case $|\lambda|$ being even.

For $|\lambda|$ being odd, we need to consider *nearly even ordered set partitions* $(B_1, B_2, \ldots, B_k)$ of M, i.e., partitions wherein each block B_i $(1 \leqslant i < k)$ has even size while the last block B_k has odd size. The construction of the involution on nearly even ordered set partitions for proving Theorem 2.2 (2) is almost identical to the previous case and will be omitted. This completes the proof of the theorem. □

Remark 2.3. *The involution ι in the special case $M = [n] := \{1, 2, \ldots, n\}$ was constructed by Sagan [29], where he extended ι to deal with some generalized Euler numbers.*

2.1. **Some q-notations and q-Euler numbers.** Before proceeding, we need to recall several q-notations to be utilized in subsequent discussions. As usual, the *q-shifted factorial* $(t;q)_n$ is defined by $(t;q)_n := \prod_{i=0}^{n-1}(1-tq^i)$ for any positive integer n and $(t;q)_0 = 1$. Then the *q-binomial numbers* $\begin{bmatrix} n \\ k \end{bmatrix}_q$ are defined as

$$\begin{bmatrix} n \\ k \end{bmatrix}_q := \frac{(q;q)_n}{(q;q)_{n-k}(q;q)_k} \qquad \text{for} \quad 0 \leqslant k \leqslant n,$$

and $\begin{bmatrix} n \\ k \end{bmatrix}_q = 0$ if $k < 0$ or $k > n$. It is known [31, Proposition 1.3.17] that the q-binomial coefficient has the following combinatorial interpretation

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{(\mathcal{X},\mathcal{Y})} q^{\mathsf{inv}(\mathcal{X},\mathcal{Y})}, \tag{2.3}$$

where the sum is over all set compositions $(\mathcal{X}, \mathcal{Y})$ of $[n]$ such that $|\mathcal{X}| = k$ and

$$\mathsf{inv}(\mathcal{X}, \mathcal{Y}) := |\{(x, y) \in \mathcal{X} \times \mathcal{Y} : x > y\}|.$$

The *q-sine* $\sin_q(x)$ and *q-cosine* $\cos_q(x)$ are defined by

$$\sin_q(x) := \sum_{n\geqslant 0} (-1)^n \frac{x^{2n+1}}{(q;q)_{2n+1}} \quad \text{and} \quad \cos_q(x) := \sum_{n\geqslant 0} (-1)^n \frac{x^{2n}}{(q;q)_{2n}}.$$

The *q-tangent* and *q-secant* functions are then defined as

$$\sec_q(x) := \frac{1}{\cos_q(x)} \quad \text{and} \quad \tan_q(x) := \frac{\sin_q(x)}{\cos_q(x)}.$$

The classical *q-Euler numbers* $E_n(q)$ then occur in the expansion of $\tan_q(x) + \sec_q(x)$:

$$\tan_q(x) + \sec_q(x) = \sum_{n\geqslant 0} E_n(q) \frac{x^n}{(q;q)_n}.$$

A q-analog of André's result (see [13, 34]) asserts that

$$E_n(q) = \sum_{\pi \in \mathcal{A}_n} q^{\mathsf{inv}(\pi)}, \tag{2.4}$$

where $\mathsf{inv}(\pi) := |\{(i, j) \in [n]^2 : i < j, \pi_i > \pi_j\}|$ is the *inversion number* of a permutation $\pi \in \mathfrak{S}_n$. The q-Euler numbers $E_n(q)$ were studied by Andrews and Gessel [3], Foata [12], Foata and Han [13], Zhao–Lin–Zang [40] and Kim and Lin [21] from both the combinatorial and number-theoretic perspectives.

For an ordered set partition $\pi = (B_1, B_2, \ldots, B_k)$ of $[n]$, let

$$\mathsf{inv}(\pi) := |\{(a, b) : a > b, a \in B_i \text{ and } b \in B_j \text{ for some } i < j\}|$$

be the number of inversions of π. Recall in the proof of Theorem 2.2 that we write $\pi \vDash_2 [2n]$ if each block B_i of π has even size. Extending this notation to the odd case, we write $\pi \vDash_2 [2n+1]$ if $\pi = (B_1, B_2, \ldots, B_k)$ is a nearly even ordered set partition of $[2n+1]$, i.e., each block B_i has even size except the last block B_k, which has odd size. As the involution ι constructed in the proof of Theorem 2.2 preserves the number of inversions of ordered set partitions, we are rewarded with the following q-analogs.

Theorem 2.4. *For $n \geqslant 0$, we have*

$$\sum_{\pi \vDash_2 [2n]} (-1)^{\mathrm{bk}(\pi)} q^{\mathsf{inv}(\pi)} = (-1)^n E_{2n}(q), \tag{2.5}$$

$$\sum_{\pi \vDash_2 [2n+1]} (-1)^{\mathrm{bk}(\pi)} q^{\mathsf{inv}(\pi)} = (-1)^{n+1} E_{2n+1}(q). \tag{2.6}$$

Consequently,

$$\sum_{k=0}^{n} \begin{bmatrix} 2n \\ 2k \end{bmatrix}_q (-1)^k E_{2k}(q) = 0, \tag{2.7}$$

$$\sum_{k=0}^{n} \begin{bmatrix} 2n+1 \\ 2k \end{bmatrix}_q (-1)^k E_{2k}(q) = (-1)^n E_{2n+1}(q), \tag{2.8}$$

$$\sum_{k=0}^{n} \begin{bmatrix} 2n+1 \\ 2k+1 \end{bmatrix}_q (-1)^k E_{2k+1}(q) = 1. \tag{2.9}$$

Proof. Here (2.5) and (2.6) follow from the involution ι. As a consequence of (2.5) and (2.6), we can deduce (2.7), (2.8) and (2.9). We present only the proof of (2.9), as the remaining two identities can be established in a similar manner.

To see (2.9), observe that each element $(B_1, B_2, \ldots, B_l) \vDash_2 [2n+1]$ with $l > 1$ can be decomposed as a subset B_1 of even size and $(B_2, \ldots, B_l) \vDash_2 [2n+1] \backslash B_1$. It then follows from (2.3) and (2.6) that

$$\begin{aligned}(-1)^{n+1} E_{2n+1}(q) = \sum_{\pi \vDash_2 [2n+1]} (-1)^{\mathrm{bk}(\pi)} q^{\mathsf{inv}(\pi)} &= -1 - \sum_{k=1}^{n} \begin{bmatrix} 2n+1 \\ 2k \end{bmatrix}_q (-1)^{n-k+1} E_{2n-2k+1}(q) \\ &= -1 + \sum_{k=0}^{n-1} \begin{bmatrix} 2n+1 \\ 2k+1 \end{bmatrix}_q (-1)^k E_{2k+1}(q),\end{aligned}$$

which is equivalent to (2.9). □

Consider the *dual q-Euler numbers*

$$E_n^*(q) := \sum_{\pi \in \mathcal{A}_n^*} q^{\mathsf{inv}(\pi)},$$

where $\mathcal{A}_n^*$ denotes the set of all down-up permutations on $[n]$. Note that

$$E_{2n}^*(q) = q^{\binom{2n}{2}} E_{2n}(1/q) \quad \text{and} \quad E_{2n+1}^*(q) = E_{2n+1}(q).$$

Let $\mathcal{E}_{2n}$ be the set of all ordered set partitions $(B_1, B_2, \ldots, B_k) \vDash [2n]$ with $k \geqslant 2$ such that each block B_i has even size, except that the first block B_1 and the last block B_k have odd sizes. For example, $3|2457|18|6$ and $6|2458|137$ are two elements in $\mathcal{E}_8$. The next identity (2.10) in essence complements (2.7), (2.8) and (2.9).

Theorem 2.5. *For $n \geqslant 0$, we have*

$$\sum_{k=0}^{n-1} \begin{bmatrix} 2n \\ 2k+1 \end{bmatrix}_q (-1)^k E_{2k+1}(q) = (-1)^{n+1} E_{2n}^*(q) + 1. \tag{2.10}$$

Consequently,

$$\sum_{\pi \in \mathcal{E}_{2n}} (-1)^{\mathrm{bk}(\pi)} q^{\mathsf{inv}(\pi)} = (-1)^{n+1} E_{2n}^*(q) + 1. \tag{2.11}$$

Proof. Let $\mathcal{M}_{2n}$ be the set of all pairs (π, σ) of words such that π is increasing with odd size, σ is up-down with odd size, and the concatenation $\pi \cdot \sigma$ is a permutation in $\mathfrak{S}_{2n}$. It follows from (2.3) and (2.4) that

$$\sum_{(\pi,\sigma) \in \mathcal{M}_{2n}} (-1)^{(|\sigma|-1)/2} q^{\mathsf{inv}(\pi \cdot \sigma)} = \sum_{k=0}^{n-1} \begin{bmatrix} 2n \\ 2k+1 \end{bmatrix}_q (-1)^k E_{2k+1}(q).$$

Hence, to prove (2.10), it remains to construct an involution on $\mathcal{M}_{2n}$ to show that

$$\sum_{(\pi,\sigma) \in \mathcal{M}_{2n}} (-1)^{(|\sigma|-1)/2} q^{\mathsf{inv}(\pi \cdot \sigma)} = (-1)^{n+1} E_{2n}^*(q) + 1. \tag{2.12}$$

Given a pair $(\pi,\sigma)\in\mathcal{M}_{2n}$ with $\pi=\pi_1\cdots\pi_k$ and $\sigma=\sigma_1\cdots\sigma_l$, define $\phi(\pi,\sigma)$ to be the new pair (π^*,σ^*) constructed as follows:

- **Case 1:** If $\pi_k<\sigma_1$ and $l>1$, then remove the first two elements $\sigma_1\sigma_2$ from σ to get σ^*, and append $\sigma_1\sigma_2$ to the end of π to get π^*.
- **Case 2:** If $\pi_k>\sigma_1$ and $k>1$, then remove the last two elements $\pi_{k-1}\pi_k$ from π to get π^*, and append $\pi_{k-1}\pi_k$ to the beginning of σ to get σ^*.
- **Case 3:** Otherwise, either $\pi_k<\sigma_1$ and $l=1$ so that the concatenation $\pi\cdot\sigma$ is the identity permutation, or $\pi_k>\sigma_1$ and $k=1$ so that the concatenation $\pi\cdot\sigma$ is a down-up permutation in $\mathcal{A}^*_{2n}$. We simply set $(\pi^*,\sigma^*)=(\pi,\sigma)$.

Clearly in all three cases we have $\pi^*\cdot\sigma^*=\pi\cdot\sigma$ so in particular $\mathsf{inv}(\pi^*\cdot\sigma^*)=\mathsf{inv}(\pi\cdot\sigma)$. Let $\mathcal{M}^*_{2n}$ denote the set of all pairs in $\mathcal{M}_{2n}$ that fall into Case 3 above. It is evident that ϕ is an involution on $\mathcal{M}_{2n}$ preserving inversion numbers, fixing elements in $\mathcal{M}^*_{2n}$, and sign-reversing on $\mathcal{M}_{2n}\backslash\mathcal{M}^*_{2n}$. Thus,

$$\sum_{(\pi,\sigma)\in\mathcal{M}_{2n}}(-1)^{(|\sigma|-1)/2}q^{\mathsf{inv}(\pi\cdot\sigma)}=\sum_{(\pi,\sigma)\in\mathcal{M}^*_{2n}}(-1)^{(|\sigma|-1)/2}q^{\mathsf{inv}(\pi\cdot\sigma)}=1+(-1)^{n-1}E^*_{2n}(q),$$

as desired. This completes the proof of (2.10).

To see (2.11), observe that each element $(B_1,B_2,\ldots,B_l)\in\mathcal{E}_{2n}$ can be decomposed as a subset $B_1\subseteq[2n]$ of odd size and $(B_2,\ldots,B_l)\vDash_2[2n]\backslash B_1$. It then follows from (2.3) and (2.6) that

$$\sum_{\pi\in\mathcal{E}_{2n}}(-1)^{\mathrm{bk}(\pi)}q^{\mathsf{inv}(\pi)}=\sum_{k=0}^{n-1}\begin{bmatrix}2n\\2k+1\end{bmatrix}_q(-1)^kE_{2k+1}(q).$$

Combining with (2.10) yields (2.11). □

3. Combinatorial identities involving Stirling numbers of the second kind

Inspired by Sagan's involution ι (see Remark 2.3) on even ordered set partitions and recent work of Zhao, Lin and Zang [40], Du [10] asked whether there is any Sagan-style involution for the following known combinatorial identity:

$$\sum_{k=1}^{n}(-1)^k2^{n-k}k!S(n,k)=\begin{cases}(-1)^{m+1}E_{2m+1}, & n=2m+1 \text{ is odd},\\ 0, & n \text{ is even}.\end{cases}\tag{3.1}$$

Here $S(n,k)$ is the Stirling number of the second kind, which counts set partitions of $[n]$ into k blocks. By convention, we define $S(0,k)=\delta_{0,k}$. In fact, identity (3.1) is a consequence of the Frobenius formula [14]

$$A_n(t)=\sum_{k=1}^{n}(t-1)^{n-k}k!S(n,k)\tag{3.2}$$

and Euler's formula [11] (see also [25, Theorem 4])

$$A_n(-1)=\begin{cases}(-1)^mE_{2m+1}, & n=2m+1 \text{ is odd},\\ 0, & n \text{ is even}.\end{cases}\tag{3.3}$$

There is a secant-number companion to (3.1), which was found by Lin, Yan and Zhao [24, Corollary 3.4] in their study of permanent evaluation:

$$\sum_{k,i}\binom{n}{k}(-1)^i 2^{k-i} i! S(k,i) = \begin{cases}(-1)^m E_{2m}, & n=2m \text{ is even},\\ 0, & n \text{ is odd}.\end{cases} \tag{3.4}$$

In this section, we will present Sagan-style involutions for q-analogs of (3.1) and (3.4).

3.1. Combinatorial set-ups and q-analogs of (3.1) and (3.4) with applications.

Definition 3.1. *A word $w = w_1 w_2 \cdots w_k$ on $\mathbb{N}$ with distinct letters is called unimodal if*

$$w_1 < \cdots < w_{k-1} < w_m > w_{m+1} > \cdots > w_k$$

for some $1 \leqslant m \leqslant k$.

Let $\mathcal{U}_n$ be the set of all unimodal permutations in $\mathfrak{S}_n$. For instance, we have $\mathcal{U}_3 = \{123, 132, 231, 321\}$. It is clear that $|\mathcal{U}_n| = 2^{n-1}$. Moreover, enumerating unimodal permutations in $\mathcal{U}_n$ by inversion numbers gives $(-q;q)_{n-1}$, as was shown in [40].

Lemma 3.2. *For $n \geqslant 1$, we have*

$$\sum_{\pi\in\mathcal{U}_n} q^{\mathsf{inv}(\pi)} = (-q;q)_{n-1}. \tag{3.5}$$

Definition 3.3. *A tuple $\pi = (\pi^{(1)}, \pi^{(2)}, \ldots, \pi^{(\ell)})$ of (non-empty) unimodal words is said to be a unimodal composition of $[n]$ if the word concatenation $\pi^{(1)}\cdot\pi^{(2)}\cdots\pi^{(\ell)}$ is a permutation in $\mathfrak{S}_n$. Define*

$$\mathsf{sgn}(\pi) = (-1)^\ell \quad \text{and} \quad \mathsf{inv}(\pi) = \mathsf{inv}(\pi^{(1)}\cdot\pi^{(2)}\cdots\pi^{(\ell)}).$$

For example, $\pi = (356, 98, 2471)$ is a unimodal composition of $[9]$ with $\mathsf{sgn}(\pi) = -1$ and $\mathsf{inv}(\pi) = \mathsf{inv}(356982471) = 20$.

Let $\mathcal{UC}_n$ be the set of all unimodal compositions of $[n]$. It is clear that

$$\sum_{\pi\in\mathcal{UC}_n}\mathsf{sgn}(\pi) = \sum_{k=1}^{n}(-1)^k 2^{n-k} k! S(n,k). \tag{3.6}$$

Definition 3.4. *A modified unimodal composition of $[n]$ is a tuple $\pi = (\pi^{(0)}, \pi^{(1)}, \ldots, \pi^{(\ell)})$ of words such that either π is the permutation $n(n-1)\cdots 1$ when $\ell = 0$, or when $\ell \geqslant 1$,*

1. *$\pi^{(0)}$ is a decreasing sequence (possibly empty) and $\pi^{(i)}$ $(1 \leqslant i \leqslant \ell)$ are non-empty unimodal words;*
2. *and the word concatenation $\pi^{(0)}\cdot\pi^{(1)}\cdots\pi^{(\ell)}$ is a permutation in $\mathfrak{S}_n$.*

Define

$$\mathsf{sgn}(\pi) = (-1)^\ell \quad \text{and} \quad \mathsf{inv}(\pi) = \mathsf{inv}(\pi^{(0)}\cdot\pi^{(1)}\cdots\pi^{(\ell)}).$$

For example, $\pi = (653, 98, 2471)$ is a modified unimodal composition of $[9]$ with $\mathsf{sgn}(\pi) = 1$ and $\mathsf{inv}(\pi) = \mathsf{inv}(653982471) = 23$.

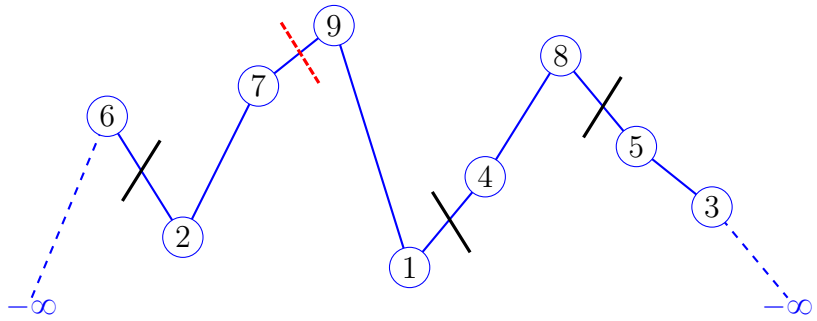


FIGURE 1. An illustration of case (1) of $\kappa : 6|2791|48|53 \mapsto 6|27|91|48|53$.

Let $\widetilde{\mathcal{UC}}_n$ be the set of all modified unimodal compositions of $[n]$. It is clear that

$$\sum_{\pi\in\widetilde{\mathcal{UC}}_n} \mathsf{sgn}(\pi) = \sum_{k,i} \binom{n}{k} (-1)^i 2^{k-i} i! S(k,i). \tag{3.7}$$

In view of (3.6) and (3.7), the following identities are q-analogs of (3.1) and (3.4).

Theorem 3.5. *For $n \geqslant 1$, we have*

$$\sum_{\pi\in\mathcal{UC}_n} \mathsf{sgn}(\pi) q^{\mathsf{inv}(\pi)} = \begin{cases} (-1)^{m+1} E_{2m+1}(q), & n = 2m+1 \text{ is odd}, \\ 0, & n \text{ is even}; \end{cases} \tag{3.8}$$

and

$$\sum_{\pi\in\widetilde{\mathcal{UC}}_n} \mathsf{sgn}(\pi) q^{\mathsf{inv}(\pi)} = \begin{cases} (-1)^{m} E_{2m}(q), & n = 2m \text{ is even}, \\ 0, & n \text{ is odd}. \end{cases} \tag{3.9}$$

Proof. It is convenient to represent an element $\pi = (\pi^{(1)}, \pi^{(2)}, \ldots, \pi^{(\ell)}) \in \mathcal{UC}_n$ by a permutation separated by bars:

$$\pi = \pi^{(1)}|\pi^{(2)}|\cdots|\pi^{(\ell)}.$$

A letter π_i of π is called a *double descent* (resp., *double ascent*, *valley*) if $\pi_{i-1} > \pi_i > \pi_{i+1}$ (resp., $\pi_{i-1} < \pi_i < \pi_{i+1}$, $\pi_{i-1} > \pi_i < \pi_{i+1}$), where we use the convention $\pi_0 = \pi_{n+1} = -\infty$. Note that a permutation is unimodal if and only if it contains no valleys. A valley π_i of π is said to be of

- *type I* if both its left and right slots receive bars in π;
- *type II* if only its left slot receives a bar in π;
- *type III* if only its right slot receives a bar in π.

For instance, in $6|2791|48|53$, the valley 2 is of type II while the valley 1 is of type III. Now we can construct an involution κ on $\mathcal{UC}_n$ to prove (3.8) according to the following cases:

(1) If π has at least one double descent/ascent, then find the leftmost double descent/ascent π_i and either delete or insert a bar in the left (resp., right) slot of π_i if π_i is a double descent (resp., double ascent). Denote the outcome composition by $\kappa(\pi)$. See Fig. 1 for an example, where the dash line in red marks the slot to which an additional bar is going to be inserted.

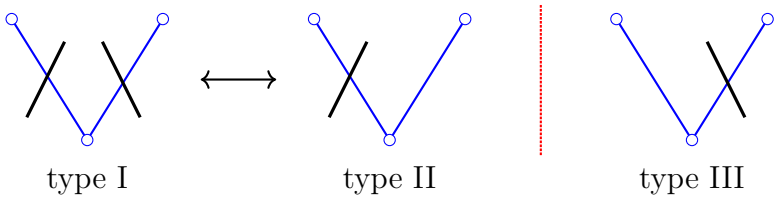


FIGURE 2. An illustration of case (2) of κ.

(2) If π has neither double descents nor double ascents (i.e., the concatenation $\pi^{(1)} \cdot \pi^{(2)} \cdots \pi^{(\ell)}$ is down-up with n odd) and contains a valley of type I/II, then find the leftmost valley of type I/II and swap the type of this valley between type I and type II. Denote the outcome composition by $\kappa(\pi)$. For example, if $\pi = 62|91|4|3|8|57$, then $\kappa(\pi) = 62|91|4|38|57$; see also the illustration in Fig. 2.
(3) If π has neither double descents nor double ascents, and all its valleys are of type III, then let $\kappa(\pi) = \pi$.

Let $\mathcal{UC}_n^*$ be the set of all $\pi \in \mathcal{UC}_n$ that fall into case (3) above. Note that $\mathcal{UC}_n^*$ is empty if n is even. It is clear that κ is an inversion number preserving involution over $\mathcal{UC}_n$, fixing elements in $\mathcal{UC}_n^*$ and sign-reversing on $\mathcal{UC}_n \backslash \mathcal{UC}_n^*$. Thus,

$$\begin{aligned}\sum_{\pi\in\mathcal{UC}_n} \mathsf{sgn}(\pi) q^{\mathsf{inv}(\pi)} &= \sum_{\pi\in\mathcal{UC}_n^*} \mathsf{sgn}(\pi) q^{\mathsf{inv}(\pi)} \\ &= \begin{cases} (-1)^{m+1} E_{2m+1}(q), & n = 2m+1 \text{ is odd}, \\ 0, & n \text{ is even}. \end{cases}\end{aligned}$$

This proves (3.8).

To show (3.9), we introduce an involution $\tilde{\kappa}$ on $\widetilde{\mathcal{UC}}_n$. Given $\pi = (\pi^{(0)}, \pi^{(1)}, \ldots, \pi^{(\ell)}) \in \widetilde{\mathcal{UC}}_n$, we consider the following cases.

(1) If $\pi^{(0)}$ is decreasing (i.e., $\pi^{(0)} \neq \varnothing$), then let $\tilde{\kappa}(\pi) = (\varnothing, \pi^{(0)}, \ldots, \pi^{(\ell)})$.
(2) If $\pi^{(0)} = \varnothing$ and $\pi^{(1)}$ is decreasing, then let $\tilde{\kappa}(\pi) = (\pi^{(1)}, \ldots, \pi^{(\ell)})$.
(3) If $\pi^{(0)} = \varnothing$ and $\pi^{(1)}$ is not decreasing, then we will remove $\pi^{(0)}$ and represent π as a barred permutation

$$\pi = \pi^{(1)}|\pi^{(2)}|\cdots|\pi^{(\ell)}.$$

Since $\pi^{(1)}$ is unimodal, the condition that $\pi^{(1)}$ is not decreasing forces $|\pi^{(1)}| > 1$ and $\pi_1 < \pi_2$. Thus, π_1 is a double ascent of π. However, we will not regard π_1 as a double ascent and apply essentially the same construction as that of κ.

(3a) If π has at least one double descent/ascent, then find the leftmost double descent/ascent π_i and either delete or insert a bar in the left (resp., right) slot of π_i if π_i is a double descent (resp., double ascent). Denote the outcome composition by $\tilde{\kappa}(\pi)$.
(3b) If π has neither double descents nor double ascents (i.e., the concatenation $\pi^{(1)} \cdot \pi^{(2)} \cdots \pi^{(\ell)}$ is up-down with n even) and contains a valley of type I/II, then find the leftmost valley of type I/II and swap the type of this valley between type I and type II. Denote the outcome composition by $\tilde{\kappa}(\pi)$.

(3c) If π has neither double descents nor double ascents, and all its valleys are of type III, then let $\tilde{\kappa}(\pi)=\pi$.

Let $\widetilde{\mathcal{UC}}_n^*$ be the set of all elements in $\widetilde{\mathcal{UC}}_n$ that fall in type (3c). Note that $\widetilde{\mathcal{UC}}_n^*$ is empty if n is odd. It is clear that $\tilde{\kappa}$ is an inversion number preserving involution on $\pi\in\widetilde{\mathcal{UC}}_n$, fixing elements in $\widetilde{\mathcal{UC}}_n^*$ and sign-reversing on $\widetilde{\mathcal{UC}}_n\backslash\widetilde{\mathcal{UC}}_n^*$. Thus,

$$\begin{aligned}\sum_{\pi\in\widetilde{\mathcal{UC}}_n}\mathsf{sgn}(\pi)q^{\mathsf{inv}(\pi)}&=\sum_{\pi\in\widetilde{\mathcal{UC}}_n^*}\mathsf{sgn}(\pi)q^{\mathsf{inv}(\pi)}\\&=\begin{cases}(-1)^mE_{2m}(q), & n=2m \text{ is even},\\ 0, & n \text{ is odd}.\end{cases}\end{aligned}$$

This completes the proof (3.9). □

As applications of Theorem 3.5, we can prove combinatorially the following identities for q-Euler numbers.

Corollary 3.6. *For $n\geqslant 1$, we have*

$$\sum_{k=0}^{n-1}\begin{bmatrix}2n\\2k+1\end{bmatrix}_q(-q;q)_{2n-2k-2}(-1)^kE_{2k+1}(q)=(-q;q)_{2n-1}, \tag{3.10}$$

$$\sum_{k=0}^{n-1}\begin{bmatrix}2n+1\\2k+1\end{bmatrix}_q(-q;q)_{2n-2k-1}(-1)^kE_{2k+1}(q)+(-1)^nE_{2n+1}(q)=(-q;q)_{2n}, \tag{3.11}$$

$$\sum_{k=0}^{n}\begin{bmatrix}2n+1\\2k\end{bmatrix}_q(-q;q)_{2n-2k}(-1)^kE_{2k}(q)=q^{\binom{2n+1}{2}}, \tag{3.12}$$

$$\sum_{k=0}^{n-1}\begin{bmatrix}2n\\2k\end{bmatrix}_q(-q;q)_{2n-2k-1}(-1)^kE_{2k}(q)+(-1)^nE_{2n}(q)=q^{\binom{2n}{2}}. \tag{3.13}$$

Proof. Each tuple $(\pi^{(1)},\pi^{(2)},\ldots,\pi^{(\ell)})\in\mathcal{UC}_{2n}$ can be constructed by first choosing $S\subseteq[2n]$, then building a unimodal permutation $\pi^{(\ell)}$ on S and finally building the remaining components $(\pi^{(1)},\pi^{(2)},\ldots,\pi^{(\ell-1)})$ on $[2n]\backslash S$. It then follows from (2.3), (3.5) and (3.8) that

$$\begin{aligned}0&=\sum_{\pi\in\mathcal{UC}_{2n}}\mathsf{sgn}(\pi)q^{\mathsf{inv}(\pi)}\\&=-\sum_{k=1}^{2n}\begin{bmatrix}2n\\k\end{bmatrix}_q(-q;q)_{k-1}\sum_{\pi\in\mathcal{UC}_{2n-k}}\mathsf{sgn}(\pi)q^{\mathsf{inv}(\pi)}\\&=-\sum_{k=0}^{n-1}\begin{bmatrix}2n\\2k+1\end{bmatrix}_q(-q;q)_{2k}\sum_{\pi\in\mathcal{UC}_{2n-2k-1}}\mathsf{sgn}(\pi)q^{\mathsf{inv}(\pi)}-(-q;q)_{2n-1}\\&=\sum_{k=0}^{n-1}\begin{bmatrix}2n\\2k+1\end{bmatrix}_q(-q;q)_{2k}(-1)^{n-k+1}E_{2n-2k-1}(q)-(-q;q)_{2n-1},\end{aligned}$$

which becomes (3.10) after setting $k\leftarrow n-k-1$ in the summation.

In the same fashion, we can deduce (3.11) from (3.8), and deduce (3.12), (3.13) from (3.9). The details will be left to the interested readers. □

Remark 3.7. *As a q-analog of* (1.1)*, identity* (3.10) *was proved in [40] using the principle of inclusion-exclusion. Identity* (3.11) *was established by Andrews and Gessel [3] via generating function manipulation. Identity* (3.12) *was proved in [40], while* (3.13) *appears to be new. All these four identities in Corollary 3.6 can also be proved combinatorially using involutions on permutation pairs*[1] *introduced by Kim and Lin [21]. Here we provide an alternative combinatorial approach based on Theorem 3.5.*

4. Combinatorial identities involving Springer numbers

The *derivative polynomials* $P_n(x)$ and $Q_n(x)$ for tangent and secant functions can be defined as:

$$\frac{\mathrm{d}^n}{\mathrm{d}\theta^n}\tan(\theta) = P_n(\tan(\theta)) \quad \text{and} \quad \frac{\mathrm{d}^n}{\mathrm{d}\theta^n}\sec(\theta) = Q_n(\tan(\theta))\sec(\theta).$$

Carlitz and Scoville [8] (see also [17] for a proof via Taylor's theorem) showed that

$$\sum_{n\geqslant 0} P_n(x)\frac{z^n}{n!} = \frac{x+\tan(z)}{1-x\tan(z)} \quad \text{and} \quad \sum_{n\geqslant 0} Q_n(x)\frac{z^n}{n!} = \frac{\sec(z)}{1-x\tan(z)}.$$

Han, Li, Ma, Yeh and Yeh [17, Theorem 3] proved analytically the following combinatorial identities linking $P_n(x)$ and $Q_n(x)$:

$$Q_{2n}(x) = (-1)^n + \sum_{k=0}^{n-1}\binom{2n}{2k+1}(-1)^k P_{2n-2k-1}(x), \tag{4.1}$$

$$Q_{2n+1}(x) = \sum_{k=0}^{n}\binom{2n+1}{2k+1}(-1)^k P_{2n-2k}(x). \tag{4.2}$$

Hoffman [19] noticed that

$$P_n(1) = 2^n E_n \quad \text{and} \quad Q_n(1) = S_n,$$

where S_n are the *Springer numbers* [30] with generating function

$$\sum_{n\geqslant 0} S_n\frac{z^n}{n!} = \frac{1}{\cos(z)-\sin(z)}.$$

Therefore, (4.1) and (4.2) specialize respectively to the following identities relating Springer numbers with Euler numbers:

$$S_{2n} = (-1)^n + \sum_{k=0}^{n-1}\binom{2n}{2k+1}(-1)^k 2^{2n-2k-1}E_{2n-2k-1}, \tag{4.3}$$

$$S_{2n+1} = \sum_{k=0}^{n}\binom{2n+1}{2k+1}(-1)^k 2^{2n-2k}E_{2n-2k}. \tag{4.4}$$

The purpose of this section is to provide combinatorial proofs of (4.1) and (4.2).

[1]Here a *permutation pair* refers to a pair (π,σ) of words such that π is unimodal, σ is down-up, and the concatenation $\pi\cdot\sigma$ is a permutation in $\mathfrak{S}_n$.

Definition 4.1. *A signed permutation of* $[n]$ *is a word of the form* $\pm\pi_1 \pm \pi_2 \cdots \pm \pi_n$ *with* $\pi_1\pi_2\cdots\pi_n \in \mathfrak{S}_n$. *Let* $\mathfrak{S}_n^{\pm}$ *be the set of all signed permutations of* $[n]$. *For convenience, we denote* $-k$ *by* $\bar{k}$ *for every integer* k. *A signed permutation* $\pi = \pi_1\cdots\pi_n \in \mathfrak{S}_n^{\pm}$ *is called a snake if it satisfies the down-up condition*

$$\pi_1 > \pi_2 < \pi_3 > \pi_4 < \cdots .$$

Let $\mathcal{S}_n \subset \mathfrak{S}_n^{\pm}$ *denote the set of snakes of length* n, *and let* $\mathcal{S}_n^0 \subset \mathcal{S}_n$ *be the subset of snakes* π *with* $\pi_1 > 0$. *For instance,* $\bar{1}32\bar{4}5$ *is a snake in* $\mathcal{S}_5\backslash\mathcal{S}_5^0$, *while* $2\bar{3}1\bar{4}5$ *is a snake in* $\mathcal{S}_5^0$.

The sets $\mathcal{S}_n$ and $\mathcal{S}_n^0$ were introduced by Arnol'd [4], under the names "β-snakes" and "snakes of type B", respectively. Following Josuat-Vergès [20], we adopt the conventions for the values of π_0 and π_{n+1}:

- If $\pi \in \mathcal{S}_n$, then define $\pi_0 = -1$ and $\pi_{n+1} = (-1)^n$;
- If $\pi \in \mathcal{S}_n^0$, then define $\pi_0 = 0$ and $\pi_{n+1} = (-1)^n$.

As the above conventions are inconsistent despite of the obvious set inclusion $\mathcal{S}_n^0 \subset \mathcal{S}_n$, we think of these two sets to be disjoint to avoid ambiguity in the sequel.

Definition 4.2. *Let* π *be a snake in* $\mathcal{S}_n$ *or* $\mathcal{S}_n^0$ *with* π_0 *and* π_{n+1} *defined as above. We define the statistic* $\mathrm{cs}(\pi)$ *as the number of changes of sign when we scan the sequence* $\pi_0, \pi_1, \ldots, \pi_{n+1}$, *i.e.,*

$$\mathrm{cs}(\pi) = |\{i : 0 \leqslant i \leqslant n,\ \pi_i\pi_{i+1} < 0\}|.$$

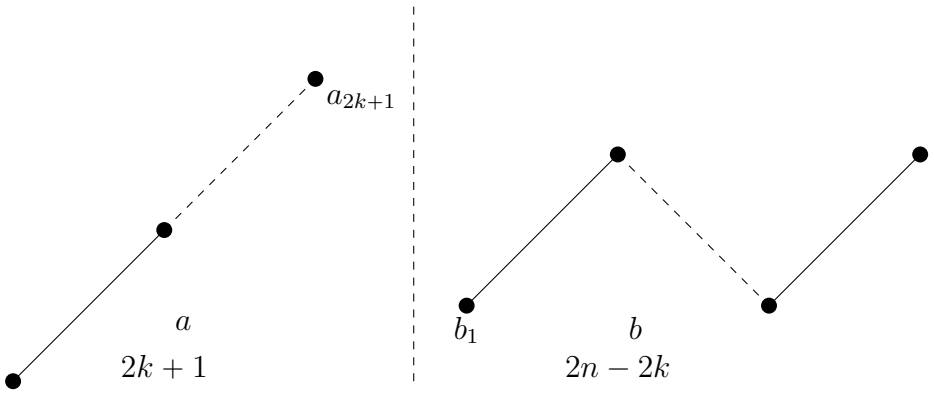
Our combinatorial proofs of (4.1) and (4.2) are based on Josuat-Vergès' interpretations [20, Theorem 3.4] of $P_n(x)$ and $Q_n(x)$:

$$P_n(x) = \sum_{\pi\in\mathcal{S}_n} x^{\mathrm{cs}(\pi)} \quad \text{and} \quad Q_n(x) = \sum_{\pi\in\mathcal{S}_n^0} x^{\mathrm{cs}(\pi)}. \tag{4.5}$$

We shall only present the combinatorial proof of (4.2), since the same idea is applicable to (4.1) as well.

Combinatorial proof of (4.2). Let $\mathcal{S}(n,k)$ denote the set of all pairs (a,b) of words such that

- $a = a_1a_2\ldots a_{2k+1}$ is an increasing sequence of length $2k+1$ with $a_1 > 0$;
- $b = b_1b_2\ldots b_{2n-2k}$ is an up-down sequence of length $2n-2k$;
- and the concatenation $a\cdot b$ is a signed permutation in $\mathfrak{S}_{2n+1}^{\pm}$.



For each pair $(a,b)\in\mathcal{S}(n,k)$, we define

$$\mathrm{cs}(a,b)=\#\{i:0\leqslant i\leqslant 2n-2k, b_ib_{i+1}<0\},$$

where we use the convention $b_0=1$ and $b_{2n-2k+1}=-1$. Note that $\mathrm{cs}(a,\varnothing)=1$ for any $(a,\varnothing)\in\mathcal{S}(n,n)$. It follows from (4.5) and the obvious one-to-one correspondence $\pi_1\pi_2\cdots\pi_n\mapsto\bar{\pi}_1\bar{\pi}_2\cdots\bar{\pi}_n$ that

$$\sum_{(a,b)\in\bigcup_{0\leqslant k\leqslant n}\mathcal{S}(n,k)}(-1)^{\frac{|a|-1}{2}}x^{\mathrm{cs}(a,b)}=\sum_{k=0}^{n}(-1)^k\sum_{(a,b)\in\mathcal{S}(n,k)}x^{\mathrm{cs}(a,b)}=\sum_{k=0}^{n}\binom{2n+1}{2k+1}(-1)^kP_{2n-2k}(x).$$

It remains to construct an involution ϕ on $\bigcup_{0\leqslant k\leqslant n}\mathcal{S}(n,k)$ to show that

$$\sum_{(a,b)\in\bigcup_{0\leqslant k\leqslant n}\mathcal{S}(n,k)}(-1)^{\frac{|a|-1}{2}}x^{\mathrm{cs}(a,b)}=Q_{2n+1}(x). \tag{4.6}$$

For a pair $(a,b)\in\mathcal{S}(n,k)$, define $\phi(a,b)$ as follows:

- **Case 1:** If $a_{2k+1}<b_1$, then move the first two elements b_1b_2 from b to the end of a.
- **Case 2:** If $k>0$, and either $a_{2k+1}>b_1$ or $b=\varnothing$, then move the last two elements $a_{2k}a_{2k+1}$ from a to the beginning of b.
- **Case 3:** If $k=0$, and either $a_{2k+1}>b_1$ or $b=\varnothing$, then let $\phi(a,b)=(a,b)$.

For example, if $(a,b)=(147,\bar{6}\bar{2}5\bar{3})\in\mathcal{S}(3,1)$, then $\phi(a,b)=(1,47\bar{6}\bar{2}\bar{5}3)\in\mathcal{S}(3,0)$.

Let $\mathcal{S}^*(n,0):=\{(a,b)\in\mathcal{S}(n,0):\text{either } a_1>b_1 \text{ or } b=\varnothing\}$. It is clear that ϕ is a sign-reversing involution on $\big(\bigcup_{0\leqslant k\leqslant n}\mathcal{S}(n,k)\big)\backslash\mathcal{S}^*(n,0)$ that fixes elements in $\mathcal{S}^*(n,0)$. Furthermore, ϕ preserves the statistic "cs", i.e., $\mathrm{cs}(a,b)=\mathrm{cs}(\phi(a,b))$. Thus,

$$\sum_{(a,b)\in\bigcup_{0\leqslant k\leqslant n}\mathcal{S}(n,k)}(-1)^{\frac{|a|-1}{2}}x^{\mathrm{cs}(a,b)}=\sum_{(a,b)\in\mathcal{S}^*(n,0)}x^{\mathrm{cs}(a,b)}=\sum_{\pi\in\mathcal{S}^0_{2n+1}}x^{\mathrm{cs}(\pi)}=Q_{2n+1}(x),$$

which proves (4.6). This completes the proof of (4.2). □

5. On combinatorics of Hirschhorn-like identities

In studying some congruences modulo 60 for secant numbers, Hirschhorn [18] proved the following two combinatorial identities using generating functions:

$$\sum_{k=0}^{n}(-4)^k\binom{4n}{4k}E_{4n-4k}=1 \tag{5.1}$$

and

$$\sum_{k=0}^{n}(-4)^k\binom{4n+2}{4k}E_{4n+2-4k}=1. \tag{5.2}$$

The purpose of this section is to provide combinatorial proofs for q-analogs of (5.1) and (5.2) (see Theorems 5.1 and 5.3), and the following pair of tangent-number companions:

$$\sum_{k=0}^{n}\binom{4n+1}{4k}(-4)^{k-n}E_{4n-4k+1}=1, \tag{5.3}$$

$$\sum_{k=0}^{n}(-1)^{n-k}\binom{4n+3}{4k}2^{2k-2n-1}E_{4n-4k+3}=1. \tag{5.4}$$

It is worth noting that our combinatorial approach has rewarded us with **two** distinct q-analogs for both (5.3) and (5.4) (see Theorems 5.4, 5.5, 5.6, and 5.8), a phenomenon we view as another indication that such an approach is more effective at uncovering the nature of the original identities and fostering novel discoveries.

We begin with the following q-analog of (5.1).

Theorem 5.1. *For $n \geqslant 0$, we have*

$$\sum_{k=0}^{n}\sum_{i=0}^{2k}\begin{bmatrix}4n\\4k\end{bmatrix}_q\begin{bmatrix}4k\\2i\end{bmatrix}_q(-1)^i E_{4n-4k}(q)=1. \tag{5.5}$$

To see that (5.1) is indeed a q-analog of (5.1), we need the following simple combinatorial identity.

Lemma 5.2. *For a fixed integer $n \geqslant 0$, we have*

$$\sum_{k=0}^{n}\binom{2n}{2k}(-1)^k=\begin{cases}0, & \text{if } n \text{ is odd},\\ (-1)^{\frac{n}{2}}2^n, & \text{if } n \text{ is even}.\end{cases} \tag{5.6}$$

Proof. For a positive integer n, define the blocks $S_i=\{2i-1,2i\}$ for $1\leqslant i\leqslant n$, and hence $[2n]=\bigcup_{i=1}^{n}S_i$. Let $\mathcal{S}_n:=\{S_1,\ldots,S_n\}$. For a given subset $A\subseteq[2n]$, define the families $\mathcal{S}_k^A=\{S\in\mathcal{S}_n : |A\bigcap S|=k\}$ for $k=0,1,2$. For example, if $n=5$ and $A=\{1,5,6,8\}$, then

- $\mathcal{S}_0^A=\{S_2,S_5\}=\{\{3,4\},\{9,10\}\}$,
- $\mathcal{S}_1^A=\{S_1,S_4\}=\{\{1,2\},\{7,8\}\}$,
- $\mathcal{S}_2^A=\{S_3\}=\{\{5,6\}\}$.

Note that $\mathcal{S}_0^A$, $\mathcal{S}_1^A$ and $\mathcal{S}_2^A$ always form a partition of $\mathcal{S}_n$ for any A.

We aim to introduce an involution φ on $\mathcal{E}_n:=\{A\subseteq[2n] : |A| \text{ even}\}$ to prove (5.6). Let $\mathcal{F}_n:=\{A\in\mathcal{E}_n : \mathcal{S}_1^A=\mathcal{S}_n\}$. Note that for every A contained in $\mathcal{F}_n$, we must have $|A|=|\mathcal{S}_1^A|=|\mathcal{S}_n|=n$, so in particular $\mathcal{F}_n=\varnothing$ if n is odd, and $|\mathcal{F}_n|=2^n$ if n is even. If $A\in\mathcal{F}_n$, then let $\varphi(A)=A$. If $A\in\mathcal{E}_n\backslash\mathcal{F}_n$, then we denote by $K(A)$ the block in $\mathcal{S}_n\backslash\mathcal{S}_1^A$ with the smallest subindex and define

$$\varphi(A)=\begin{cases}A\backslash K(A), & \text{if } K(A)\subseteq A,\\ A\bigcup K(A), & \text{otherwise.}\end{cases}$$

Continuing with the previous example where $n=5$ and $A=\{1,5,6,8\}$, we have $\mathcal{S}_n\backslash\mathcal{S}_1^A=\{S_2,S_3,S_5\}$ and so $K(A)=S_2=\{3,4\}$. Thus, $\varphi(A)=A\bigcup K(A)=\{1,3,4,5,6,8\}$. It is plain to see that $\mathcal{S}_1^A=\mathcal{S}_1^{\varphi(A)}$, rendering φ a well-defined involution on $\mathcal{E}_n$. Moreover, if we define the sign of $A\in\mathcal{E}_n$ to be $(-1)^{|A|/2}$, then φ is sign-reversing on $\mathcal{E}_n\backslash\mathcal{F}_n$ and fixes elements in $\mathcal{F}_n$. By this involution, we have

$$\sum_{k=0}^{n}\binom{2n}{2k}(-1)^k=\sum_{A\in\mathcal{E}_n}(-1)^{|A|/2}=\sum_{A\in\mathcal{F}_n}(-1)^{|A|/2}=\begin{cases}0, & \text{if } n \text{ is odd},\\ (-1)^{\frac{n}{2}}2^n, & \text{if } n \text{ is even}.\end{cases}$$

The proof of the lemma is complete. □

Next we prove Theorem 5.1 combinatorially, which together with our involutive proof of Lemma 5.2 leads to a combinatorial proof of Hirschhorn's identity (5.1).

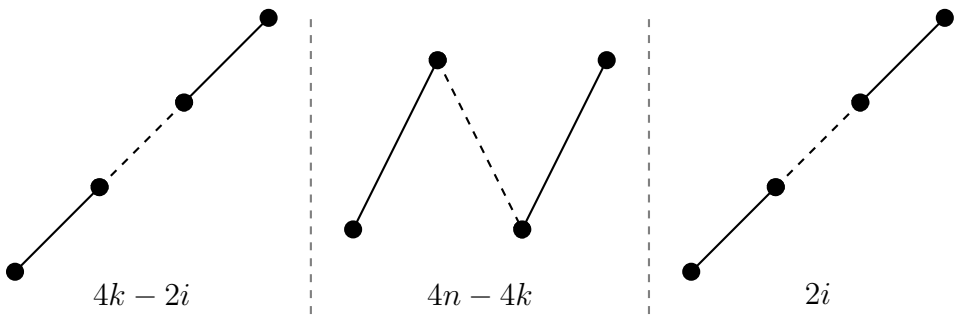


FIGURE 3. An illustration of triples in $\mathcal{T}(n,k,i)$.

Proof of Theorem 5.1. Let $\mathcal{T}(n,k,i)$ denote the set of all triples (a,b,c) of words, where

- $a = a_1a_2\dots a_{4k-2i}$ and $c = c_1c_2\dots c_{2i}$ are increasing sequences of lengths $4k-2i$ and $2i$, respectively;
- $b = b_1b_2\dots b_{4n-4k}$ is an up-down sequence of length $4n-4k$;
- and the word concatenation $a\cdot b\cdot c$ is a permutation in $\mathfrak{S}_{4n}$.

For instance, $(1468, 2735, \varnothing)$ is a triple in $\mathcal{T}(2,1,0)$; see Fig. 3 for an illustration of such triples.

For a triple $(a,b,c)\in\mathcal{T}(n,k,i)$, let $\mathsf{inv}(a,b,c) := \mathsf{inv}(a\cdot b\cdot c)$ be the *inversion number* of (a,b,c). By (2.3) and (2.4), we have

$$\sum_{(a,b,c)\in\mathcal{T}(n,k,i)} q^{\mathsf{inv}(a,b,c)} = \begin{bmatrix} 4n \\ 4k-2i \end{bmatrix}_q \begin{bmatrix} 4n-4k+2i \\ 2i \end{bmatrix}_q E_{4n-4k}(q) = \begin{bmatrix} 4n \\ 4k \end{bmatrix}_q \begin{bmatrix} 4k \\ 2i \end{bmatrix}_q E_{4n-4k}(q),$$

and so

$$\sum_{k=0}^{n}\sum_{i=0}^{2k}\sum_{(a,b,c)\in\mathcal{T}(n,k,i)} (-1)^i q^{\mathsf{inv}(a,b,c)} = \sum_{k=0}^{n}\sum_{i=0}^{2k}(-1)^i \begin{bmatrix} 4n \\ 4k \end{bmatrix}_q \begin{bmatrix} 4k \\ 2i \end{bmatrix}_q E_{4n-4k}(q).$$

Hence, it remains to show that

$$\sum_{k=0}^{n}\sum_{i=0}^{2k}\sum_{(a,b,c)\in\mathcal{T}(n,k,i)} (-1)^i q^{\mathsf{inv}(a,b,c)} = 1$$

by constructing a sign-reversing involution ψ on $\mathcal{T}_n := \bigcup_{k,i}\mathcal{T}(n,k,i)$.

Given $(a,b,c)\in\mathcal{T}(n,k,i)$, define $\psi(a,b,c)$ to be the triple obtained as follows:

- **Case 1:** b is non-empty.
  - i) If a is non-empty with $a_{4k-2i} > b_1$, then move $a_{4k-2i-1}a_{4k-2i}$ from a to the beginning of b; otherwise, move b_1b_2 from b to the end of a, this includes the case when $a=\varnothing$.
  - ii) If c is non-empty with $c_1 < b_{4n-4k}$, then move c_1c_2 from c to the end of b; otherwise, move $b_{4n-4k-1}b_{4n-4k}$ from b to the beginning of c, this includes the case when $c=\varnothing$.
- **Case 2:** b is empty and $a_{4k-2i} > c_1$. Move $a_{4k-2i-1}a_{4k-2i}$ from a to the beginning of b and then move c_1c_2 from c to the end of b. I.e., the middle word in the image triple $\psi(a,b,c)$ is precisely $a_{4k-2i-1}a_{4k-2i}c_1c_2$.

- **Case 3:** those triples not covered by Cases 1 and 2 above, i.e., b is empty and either $a_{4k-2i} < c_1$, or a is empty, or c is empty. Let $\psi(a,b,c) = (a,b,c)$ (i.e., no operation is performed).

For example, if $(a,b,c) = (2\,4, 1\,5\,3\,11\,8\,12\,6\,7, 9\,10) \in \mathcal{T}(3,1,1)$, then we have $\psi(a,b,c) = (\varnothing, \mathbf{2}\,\mathbf{4}\,1\,5\,3\,11\,8\,12, \mathbf{6}\,\mathbf{7}\,9\,10) \in \mathcal{T}(3,1,2)$.

Denote by $\mathcal{T}_n^*$ the set of triples in $\mathcal{T}_n$ that fall in Case 3 above. It is clear that ψ is an involution on $\mathcal{T}_n$ preserving inversion numbers, fixing elements in $\mathcal{T}_n^*$ and sign-reversing on $\mathcal{T}_n \backslash \mathcal{T}_n^*$. Thus,

$$\sum_{k=0}^{n}\sum_{i=0}^{2k}\sum_{(a,b,c)\in\mathcal{T}(n,k,i)}(-1)^i q^{\mathsf{inv}(a,b,c)} = \sum_{(a,\varnothing,c)\in\mathcal{T}_n^*}(-1)^{\frac{|c|}{2}} = \sum_{i=0}^{2n}(-1)^i = 1,$$

as desired. This completes the proof of the theorem. □

Using similar approach as in Theorem 5.1, we can obtain the following q-identity which is a q-analog of (5.2) in view of Lemma 5.2.

Theorem 5.3. *For $n \geqslant 0$, we have*

$$\sum_{k=0}^{n}\sum_{i=0}^{2k}\begin{bmatrix}4n+2\\4k\end{bmatrix}_q\begin{bmatrix}4k\\2i\end{bmatrix}_q(-1)^i E_{4n-4k+2}(q) = 1. \tag{5.7}$$

Proof. The proof is essentially the same as that of Theorem 5.1 and is left to the interested reader. □

Next, we present four q-analogs of (5.3) and (5.4), two for each identity. The main ideas are similar to those in the proof of Theorem 5.1, although the details for the second q-analog of (5.3) are more involved.

Theorem 5.4. *For $n \geqslant 0$, we have*

$$\sum_{k=0}^{n}\sum_{i=0}^{2k}\begin{bmatrix}4n+1\\4k\end{bmatrix}_q\begin{bmatrix}4k\\2i\end{bmatrix}_q(-1)^i q^{i(2i-1)}E_{4n-4k+1}(q) = \sum_{i=0}^{2n}\begin{bmatrix}4n\\2i\end{bmatrix}_q(-1)^i q^{i(2i+1)}. \tag{5.8}$$

Consequently, (5.3) holds true.

Proof. Let $\mathcal{P}(n,k,i)$ denote the set of triples (a,b,c) of words such that

- $a = a_1a_2\ldots a_{4k-2i}$ (resp., $c = c_1c_2\ldots c_{2i}$) is an increasing (resp., decreasing) sequence of length $4k-2i$ (resp., $2i$);
- $b = b_1b_2\ldots b_{4n-4k+1}$ is an up-down sequence of length $4n-4k+1$,
- and the word concatenation $a\cdot b\cdot c$ is a permutation in $\mathfrak{S}_{4n+1}$.

See Fig. 4 for an illustration of such triples. As before, let $\mathsf{inv}(a,b,c) := \mathsf{inv}(a\cdot b\cdot c)$ be the *inversion number* of a triple $(a,b,c)\in\mathcal{P}(n,k,i)$. By (2.3) and (2.4), we have

$$\sum_{(a,b,c)\in\mathcal{P}(n,k,i)} q^{\mathsf{inv}(a,b,c)} = \begin{bmatrix}4n+1\\4k\end{bmatrix}_q\begin{bmatrix}4k\\2i\end{bmatrix}_q q^{i(2i-1)}E_{4n-4k+1}(q)$$

and so

$$\sum_{k=0}^{n}\sum_{i=0}^{2k}(-1)^i\sum_{(a,b,c)\in\mathcal{P}(n,k,i)} q^{\mathsf{inv}(a,b,c)} = \sum_{k=0}^{n}\sum_{i=0}^{2k}(-1)^i\begin{bmatrix}4n+1\\4k\end{bmatrix}_q\begin{bmatrix}4k\\2i\end{bmatrix}_q q^{i(2i-1)}E_{4n-4k+1}(q).$$

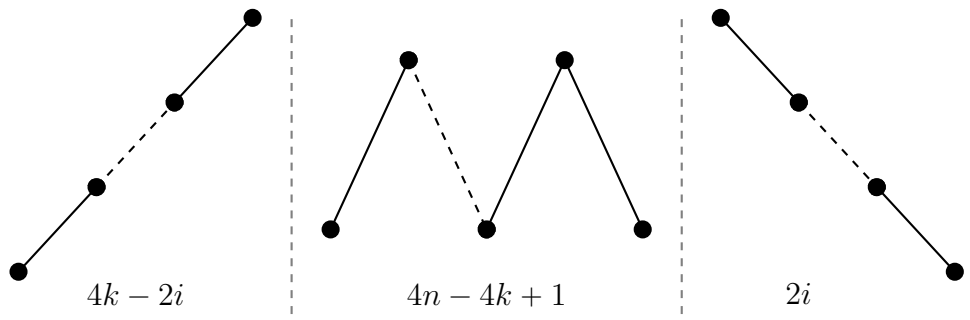


FIGURE 4. An illustration of triples in $\mathcal{P}(n,k,i)$.

Hence, it remains to show that

$$\sum_{k=0}^{n}\sum_{i=0}^{2k}\sum_{(a,b,c)\in\mathcal{P}(n,k,i)}(-1)^i q^{\mathsf{inv}(a,b,c)} = \sum_{i=0}^{2n}\begin{bmatrix}4n\\2i\end{bmatrix}_q (-1)^i q^{i(2i+1)}$$

by constructing a sign-reversing involution ϑ on $\mathcal{P}_n := \bigcup_{k,i}\mathcal{P}(n,k,i)$.

Given $(a,b,c)\in\mathcal{P}(n,k,i)$, define $\vartheta(a,b,c)$ to be the new triple $(\tilde{a},\tilde{b},\tilde{c})$ obtained as follows:

- **Case 1:** $k\neq n$ (i.e., $|b|\geqslant 5$).
  - i) If a is non-empty with $a_{4k-2i} > b_1$, then move $a_{4k-2i-1}a_{4k-2i}$ from a to the beginning of b; otherwise, move b_1b_2 from b to the end of a, this includes the case when $a=\varnothing$.
  - ii) If c is non-empty with $c_1 > b_{4n-4k+1}$, then move c_1c_2 from c to the end of b; otherwise, move $b_{4n-4k}b_{4n-4k+1}$ from b to the beginning of c, this includes the case when $c=\varnothing$.
- **Case 2:** $k=n$ (i.e., $|b|=1$). We further distinguish four cases and use the convention that $a_{4k-2i}=-\infty$ when a is empty and $c_1=-\infty$ when c is empty, for the sake of convenience.
  - (a) $a_{4k-2i} > b_1 > c_1$. Set $(\tilde{a},\tilde{b},\tilde{c}) = (a_1\ldots a_{4k-2i-2}, a_{4k-2i-1}, a_{4k-2i}b_1c_1\ldots c_{2i})$.
  - (b) $a_{4k-2i} < b_1 < c_1$. Set $(\tilde{a},\tilde{b},\tilde{c}) = (a_1\ldots a_{4k-2i}b_1c_1, c_2, c_3\ldots a_{2i})$.
  - (c) $a_{4k-2i} > b_1 < c_1$. Set $(\tilde{a},\tilde{b},\tilde{c}) = (a_1\ldots a_{4k-2i-2}, a_{4k-2i-1}a_{4k-2i}b_1c_1c_2, c_3\ldots c_{2i})$.
  - (d) $a_{4k-2i} < b_1 > c_1$. Set $(\tilde{a},\tilde{b},\tilde{c}) = (a,b,c)$.

Denote by $\mathcal{P}_n^*$ the set of triples in $\mathcal{P}_n$ that fall in Case 2-(d) above. Note that $\mathcal{P}_n^*$ is in an obvious bijection with unimodal permutations in $\mathfrak{S}_{4n+1}$ whose largest letter $4n+1$occupies an odd position. One can verify that ϑ is an involution on $\mathcal{P}_n$ preserving inversion numbers, fixing elements in $\mathcal{P}_n^*$ and sign-reversing on $\mathcal{P}_n\backslash\mathcal{P}_n^*$. Thus,

$$\sum_{k=0}^{n}\sum_{i=0}^{2k}\sum_{(a,b,c)\in\mathcal{P}(n,k,i)}(-1)^i q^{\mathsf{inv}(a,b,c)} = \sum_{(a,b,c)\in\mathcal{P}_n^*}(-1)^{|c|/2} q^{\mathsf{inv}(a,b,c)} = \sum_{i=0}^{2n}\begin{bmatrix}4n\\2i\end{bmatrix}_q (-1)^i q^{i(2i+1)},$$

as claimed. This completes the proof of (5.8).

Finally, applying Lemma 5.2 to the $q=1$ version of (5.8) yields

$$\sum_{k=0}^{n}\binom{4n+1}{4k}(-4)^k E_{4n-4k+1} = (-4)^n,$$

which is equivalent to (5.3). This completes the proof of the theorem. □

Theorem 5.5. *For $n \geqslant 0$, we have*

$$\sum_{k=0}^{n}\sum_{i=0}^{2k}\begin{bmatrix}4n+3\\4k\end{bmatrix}_q\begin{bmatrix}4k\\2i\end{bmatrix}_q(-1)^i q^{i(2i-1)}E_{4n-4k+3}(q)=\sum_{i=0}^{2n}\begin{bmatrix}4n+2\\2i+1\end{bmatrix}_q(-1)^i q^{(i+1)(2i+1)}. \tag{5.9}$$

Consequently, (5.4) holds ture.

Proof. The proof of (5.9) is similar to that of (5.8) and will be left to the interested reader, for the sake of brevity.

Setting $q=1$ in (5.9) and then applying Lemma 5.2 yields

$$\begin{aligned}\sum_{k=0}^{n}(-4)^k\binom{4n+3}{4k}E_{4n-4k+3}&=\sum_{i=0}^{2n}\binom{4n+2}{2i+1}(-1)^i\\&=\sum_{i=0}^{2n}\left(\binom{4n+1}{2i}+\binom{4n+1}{2i+1}\right)(-1)^i\\&=\sum_{i=0}^{2n}\left(\binom{4n}{2i}+\binom{4n}{2i-1}+\binom{4n}{2i+1}+\binom{4n}{2i}\right)(-1)^i\\&=2\sum_{i=0}^{2n}(-1)^i\binom{4n}{2i}=2\cdot(-4)^n,\end{aligned}$$

which is equivalent to (5.4). □

The next two theorems provide alternative q-analogs of (5.3) and (5.4).

Theorem 5.6. *For $n \geqslant 0$, we have*

$$\sum_{k=0}^{n}\sum_{i=0}^{2k}\begin{bmatrix}4n+1\\4k\end{bmatrix}_q\begin{bmatrix}4k\\2i\end{bmatrix}_q(-1)^i E_{4n-4k+1}(q)=\sum_{i=0}^{2n}\begin{bmatrix}4n+1\\2i\end{bmatrix}_q(-1)^i. \tag{5.10}$$

Remark 5.7. *Since*

$$\sum_{i=0}^{2n}\begin{bmatrix}4n+1\\2i\end{bmatrix}_q(-1)^i=\sum_{i=0}^{2n}(-1)^i\left(q^{2i}\begin{bmatrix}4n\\2i\end{bmatrix}_q+\begin{bmatrix}4n\\2i-1\end{bmatrix}_q\right)=\sum_{i=0}^{2n}\begin{bmatrix}4n\\2i\end{bmatrix}_q(-1)^i q^{2i},$$

identity (5.10) can also be written as

$$\sum_{k=0}^{n}\sum_{i=0}^{2k}\begin{bmatrix}4n+1\\4k\end{bmatrix}_q\begin{bmatrix}4k\\2i\end{bmatrix}_q(-1)^i E_{4n-4k+1}(q)=\sum_{i=0}^{2n}\begin{bmatrix}4n\\2i\end{bmatrix}_q(-1)^i q^{2i},$$

a form that closely resembles the first q-identity (5.8).

Before we proceed to prove Theorem 5.6, we need to introduce two useful operations. Given a word $w=w_1w_2\cdots w_k$ on $\mathbb{P}$ with distinct letters, let $w^r:=w_kw_{k-1}\cdots w_1$ be its reversal. We define w^{rc} as the word obtained from w^r by replacing, for each i, the i-th largest letter with its i-th smallest letter. For instance, if $w=7834196$, then $w^r=6914387$ and so $w^{rc}=6197834$. Let (a,b,c) be a triple of words whose concatenation $a\cdot b\cdot c$ is a permutation in $\mathfrak{S}_n$. Define $(a,b,c)^{rc}=(c',b',a')$, where the word concatenation $c'\cdot b'\cdot a'=$

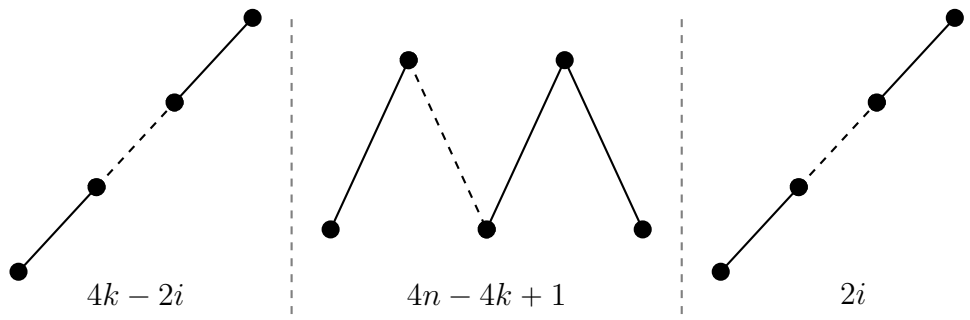


FIGURE 5. An illustration of triples in $\mathcal{Q}(n,k,i)$.

$(a\cdot b\cdot c)^{rc}$ and $|a|=|a'|$, $|b|=|b'|$, $|c|=|c'|$. For instance, if $(a,b,c)=(6,41539,287)$, then $(a,b,c)^{rc}=(328,17596,4)$. Note that $\mathsf{inv}(w)=\mathsf{inv}(w^{rc})$ and $\mathsf{inv}(a,b,c)=\mathsf{inv}((a,b,c)^{rc})$, where $\mathsf{inv}(a,b,c):=\mathsf{inv}(a\cdot b\cdot c)$.

We are ready for an involutive proof of Theorem 5.6.

Proof of Theorem 5.6. Let $\mathcal{Q}(n,k,i)$ denote the set of triples (a,b,c) of words such that

- $a=a_1a_2\ldots a_{4k-2i}$ and $c=c_1c_2\ldots c_{2i}$ are increasing sequences of length $4k-2i$ and $2i$, respectively;
- $b=b_1b_2\ldots b_{4n-4k+1}$ is an up-down sequence of length $4n-4k+1$,
- and the word concatenation $a\cdot b\cdot c$ is a permutation in $\mathfrak{S}_{4n+1}$.

See Fig. 5 for an illustration of such triples. Let $\mathcal{Q}_n:=\bigcup_{k,i}\mathcal{Q}(n,k,i)$. By (2.3) and (2.4), we have

$$\sum_{(a,b,c)\in\mathcal{Q}(n,k,i)} q^{\mathsf{inv}(a,b,c)}=\begin{bmatrix}4n+1\\4k\end{bmatrix}_q\begin{bmatrix}4k\\2i\end{bmatrix}_q E_{4n-4k+1}(q)$$

and so

$$\sum_{(a,b,c)\in\mathcal{Q}_n}(-1)^{|c|/2}q^{\mathsf{inv}(a,b,c)}=\sum_{k=0}^{n}\sum_{i=0}^{2k}(-1)^i\begin{bmatrix}4n+1\\4k\end{bmatrix}_q\begin{bmatrix}4k\\2i\end{bmatrix}_q E_{4n-4k+1}(q).$$

It remains to construct an involution κ on $\mathcal{Q}_n$ to prove that

$$\sum_{(a,b,c)\in\mathcal{Q}_n}(-1)^{|c|/2}q^{\mathsf{inv}(a,b,c)}=\sum_{i=0}^{2n}\begin{bmatrix}4n+1\\2i\end{bmatrix}_q(-1)^i.$$

Given $(a,b,c)\in\mathcal{Q}(n,k,i)$, we construct $\kappa(a,b,c)$ according to the following two cases. If $n=k$ and $a_{4k-2i}<b_1$, then define $\kappa(a,b,c)=(a,b,c)$. Otherwise, we perform the following steps to get $\kappa(a,b,c)$:

(i) If $a_{4k-2i}>b_1$, then move $a_{4k-2i-1}a_{4k-2i}$ from a to the beginning of b; otherwise, we must have $a_{4k-2i}<b_1$ (including the case when $a=\varnothing$) and $k<n$, we move b_1b_2 from b to the end of a. Let $\tilde{a}$ and $\tilde{b}$ be respectively the resulting words in place of a and b after this operation.

(ii) Note that $(\tilde{b})^{rc}$ is a down-up word of odd length. If the last letter of $(\tilde{b})^{rc}$ is greater than c_1, then move c_1c_2 from c to the end of $(\tilde{b})^{rc}$; otherwise, move the last two letters of $(\tilde{b})^{rc}$ to the beginning of c. Let b^* and c^* be respectively the resulting words in place of $(\tilde{b})^{rc}$ and c after this operation.

(iii) Define $\kappa(a,b,c)=(\tilde{a},b^*,c^*)^{rc}$.

Take $(a,b,c)=(2578,34196,\varnothing)\in\mathcal{P}(2,1,0)$ as an example, then after step (i) we get $(\tilde{a},\tilde{b},c)=(25,7834196,\varnothing)$. As $(\tilde{b})^{rc}=6197834$, we have $(\tilde{a},b^*,c^*)=(25,61978,34)$ after applying step (ii). Finally, we have $\kappa(a,b,c)=(\tilde{a},b^*,c^*)^{rc}=(67,23194,58)\in\mathcal{P}(2,1,1)$. If we apply the above three steps to $(67,23194,58)$, then we retrieve $(2578,34196,\varnothing)$:

$$(67,23194,58)\mapsto(\varnothing,6723194,58)\mapsto(\varnothing,41967,2358)\mapsto(2578,34196,\varnothing).$$

Since each step preserves the inversion number, we have $\mathsf{inv}(\kappa(a,b,c))=\mathsf{inv}(a,b,c)$, i.e., the mapping κ preserves inversion numbers. Moreover, it is routine to check step by step that κ is an involution on $\mathcal{Q}_n$ fixing elements in $\mathcal{Q}_n^*$ and sign-reversing on $\mathcal{Q}_n\backslash\mathcal{Q}_n^*$, where $\mathcal{Q}_n^*$ denotes the set of all triples $(a,b,c)\in\mathcal{Q}_n$ satisfying $|b|=1$ and the last letter of a, if any, is smaller than b_1. Thus,

$$\sum_{(a,b,c)\in\mathcal{Q}_n}(-1)^{|c|/2}q^{\mathsf{inv}(a,b,c)}=\sum_{(a,b,c)\in\mathcal{Q}_n^*}(-1)^{|c|/2}q^{\mathsf{inv}(a,b,c)}=\sum_{i=0}^{2n}\begin{bmatrix}4n+1\\2i\end{bmatrix}_q(-1)^i,$$

where the last equality follows from the interpretation (2.3) of the q-binomial coefficients. This completes the proof of the theorem. □

Similarly, we present an additional q-analog of (5.4), which is distinct from (5.9).

Theorem 5.8. *For $n\geqslant 0$, we have*

$$\sum_{k=0}^{n}\sum_{i=0}^{2k}\begin{bmatrix}4n+3\\4k\end{bmatrix}_q\begin{bmatrix}4k\\2i\end{bmatrix}_q(-1)^iE_{4n-4k+3}(q)=\sum_{i=0}^{2n+1}\begin{bmatrix}4n+3\\2i+1\end{bmatrix}_q(-1)^i. \tag{5.11}$$

Proof. The proof is essentially identical to that of Theorem 5.6 and is therefore omitted for the sake of brevity. □

5.1. **On two sequences arising from hyperbolic and trigonometric functions.** Since Hirschhorn's original proof of (5.1) and (5.2) relies on the generating function of secant numbers in terms of the hyperbolic function $\cosh(x)$, we investigate in this subsection other hyperbolic functions in the hope of unearthing further Hirschhorn-like identities. Recall the following four hyperbolic functions

$$\sinh(x)=-i\sin(ix),\ \cosh(x)=\cos(ix),\ \tanh(x)=-i\tan(ix),\quad\text{and}\quad\operatorname{sech}(x)=\sec(ix),$$

where i is the imaginary unit satisfying $i^2=-1$. The following integer sequences $\{A_n\}_{n\geqslant0}$ and $\{(-1)^{\lfloor n/2\rfloor}B_n\}_{n\geqslant0}$ are registered in the OEIS [27] as entries A009013 and A003702, respectively. Their exponential generating functions are given by

$$\sum_{n\geqslant0}A_n\frac{x^n}{n!}=\sec(x)\operatorname{sech}(x)=1+4\frac{x^4}{4!}+1104\frac{x^8}{8!}+2154304\frac{x^{12}}{12!}+15456463104\frac{x^{16}}{16!}+\cdots$$

and

$$\sum_{n\geqslant0}B_n\frac{x^n}{n!}=\sec(x)\tanh(x)=x+1\frac{x^3}{3!}+21\frac{x^5}{5!}+141\frac{x^7}{7!}+10441\frac{x^9}{9!}+183481\frac{x^{11}}{11!}+\cdots.$$

From the above definitions, we have

$$A_n = \begin{cases} \sum_{k=0}^{2m} \binom{4m}{2k}(-1)^k E_{4m-2k}E_{2k}, & \text{if } n = 4m; \\ 0, & \text{otherwise;} \end{cases}$$

and

$$B_n = \begin{cases} \sum_{k=0}^{m} \binom{2m+1}{2k}(-1)^{m+k} E_{2m-2k+1}E_{2k}, & \text{if } n = 2m+1; \\ 0, & \text{otherwise.} \end{cases}$$

The appearances of (-1) in both formulas make it unclear that $\{A_n\}_{n\geqslant 0}$ and $\{B_n\}_{n\geqslant 0}$ are nonnegative integer sequences.

Proposition 5.9. *For $n \geqslant 1$, we have*

$$A_{4n} = 2\sum_{k=1}^{n} \binom{4n-1}{4k-1} E_{4k-1}A_{4n-4k} \tag{5.12}$$

and

$$B_{2n+1} = E_{2n} + 2\sum_{k=1}^{\lfloor n/2\rfloor} \binom{2n}{4k-1} E_{4k-1}B_{2n+1-4k} \tag{5.13}$$

with initial values $A_0 = 1$ and $B_1 = 1$. Consequently, both A_{4n} and B_{2n+1} are positive integers for any $n \geqslant 0$.

Proof. Let $f(x) = \sec(x)\mathrm{sech}(x)$. Then,

$$f'(x) = f(x)(\tan(x) - \tanh(x)),$$

by which we get (5.12) after extracting the coefficients of $\frac{x^{4n-1}}{(4n-1)!}$ from both sides. Similarly, if we denote $g(x) = \sec(x)\tanh(x)$, then

$$g'(x) = \sec(x) + g(x)(\tan(x) - \tanh(x)).$$

Extracting the coefficients of $\frac{x^{2n}}{(2n)!}$ from both sides of the above equation yields (5.13). □

Corollary 5.10. *The integer A_{4n} is divisible by 4^n, and the quotient is an odd number.*

Proof. It is known (see [40]) that 2^{2n} divides $(n+1)E_{2n+1}$ for $n \geqslant 0$. Thus, 2^{n+1} divides E_{2n+1} holds for $n \geqslant 2$. By (5.12), we have

$$A_{4n} = 4\binom{4n-1}{3}A_{4n-4} + 2\sum_{k=2}^{n} \binom{4n-1}{4k-1} E_{4k-1}A_{4n-4k},$$

which can be rewritten as

$$\frac{A_{4n}}{4^n} = \binom{4n-1}{3}\frac{A_{4n-4}}{4^{n-1}} + 2\sum_{k=2}^{n} \binom{4n-1}{4k-1} \frac{E_{4k-1}}{2^{2k}}\frac{A_{4n-4k}}{4^{n-k}}. \tag{5.14}$$

We proceed to prove the result by induction on n. By the induction hypothesis, $\binom{4n-1}{3}\frac{A_{4n-4}}{4^{n-1}}$ is odd and $\frac{E_{4k-1}}{2^{2k}}\frac{A_{4n-4k}}{4^{n-k}}$ are integers for $k \geqslant 2$ (as 2^{n+1} divides E_{2n+1} for $n \geqslant 2$). It then follows from (5.14) that $\frac{A_{4n}}{4^n}$ is odd, which completes the proof by induction. □

A permutation $\pi = \pi_1\pi_2\cdots\pi_{4n} \in \mathfrak{S}_{4n}$ is called 4-*alternating* if $\pi_i > \pi_{i+1}$ precisely when $i = 4, 8, \ldots, 4(n-1)$. For example, $35681247 \in \mathfrak{S}_8$ is 4-alternating. Note that 4-alternating permutations were considered by Carlitz [6]. Let $\mathfrak{C}_{4n}$ be the set of 4-alternating permutations in $\mathfrak{S}_{4n}$. Introduce the inversion enumerator

$$a_{4n}(q) := \sum_{\pi\in\mathfrak{C}_{4n}} q^{\mathsf{inv}(\pi)}.$$

Let $\tilde{A}_{4n} = \frac{A_{4n}}{4^n}$ be the reduced sequence of A_{4n}, which is also recorded in [27, A211212]. The following result shows that 4-alternating permutations indeed provide a combinatorial interpretation for $\tilde{A}_{4n}$, and this connection between A009013 and A211212 appears to be unnoticed before.

Proposition 5.11. *For $n \geqslant 1$, we have*

$$\sum_{k=0}^{n}(-1)^k \begin{bmatrix} 4n \\ 4k \end{bmatrix} a_{4n-4k}(q) = 0. \tag{5.15}$$

Consequently,

$$1 + \sum_{n\geqslant 1} a_{4n}(q)\frac{x^{4n}}{(q;q)_{4n}} = \left(\sum_{n\geqslant 0}(-1)^n \frac{x^{4n}}{(q;q)_{4n}}\right)^{-1}. \tag{5.16}$$

In particular, $\tilde{A}_{4n} = a_{4n}(1)$.

Proof. Let $\mathcal{C}(n,k)$ denote the set of pairs (a,b) of words such that

- $a = a_1a_2\ldots a_{4k}$ is an increasing sequence of length $4k$;
- $b = b_1b_2\ldots b_{4n-4k}$ is a 4-alternating sequence of length $4n-4k$;
- and the word concatenation $a\cdot b$ is a permutation in $\mathfrak{S}_{4n}$.

For example, $(2\,5\,6\,7, 3\,4\,10\,12\,1\,8\,9\,11)$ is a pair in $\mathcal{C}(3,1)$. Let $\mathcal{C}_n := \bigcup_k \mathcal{C}(n,k)$. By (2.3), we have

$$\sum_{(a,b)\in\mathcal{C}_n}(-1)^{|a|/4}q^{\mathsf{inv}(a\cdot b)} = \sum_{k=0}^{n}(-1)^k\begin{bmatrix} 4n \\ 4k \end{bmatrix} a_{4n-4k}(q).$$

To finish the proof of (5.15), it remains to construct a sign-reversing involution on $\mathcal{C}_n$.

Given a pair $(a,b)\in\mathcal{C}(n,k)$, construct (a',b') from (a,b) as follows:

- if $a_{4k} > b_1$ or b is empty, then (a',b') is obtained by moving the last four letters from a to the beginning of b;
- otherwise, (a',b') is obtained by moving the first four letters from b to the end of a.

The mapping $(a,b)\mapsto(a',b')$ is a sign-reversing involution on $\mathcal{C}_n$ which preserves the number of inversions. This proves (5.15), from which (5.16) follows immediately.

By Lemma 5.2, we have

$$\cos(\frac{x}{\sqrt{2}})\cosh(\frac{x}{\sqrt{2}}) = \sum_{n\geqslant 0}(-1)^n\frac{x^{4n}}{(4n)!}$$

and so $1 + \sum_{n\geqslant 1} a_{4n}(1)\frac{x^{4n}}{(4n)!} = \sec(\frac{x}{\sqrt{2}})\mathrm{sech}(\frac{x}{\sqrt{2}})$. This proves that $\tilde{A}_{4n} = a_{4n}(1)$. □

Remark 5.12. *The generating function (5.16) for 4-alternating permutations can also be computed via Zhuang's generalized run theorem [41] (see also [9]).*

Consider the q-analogs of A_n and B_n as

$$A_n(q) := \begin{cases} \sum_{k=0}^{2m} \begin{bmatrix} 4m \\ 2k \end{bmatrix}_q (-1)^k E_{4m-2k}(q) E_{2k}(q), & \text{if } n = 4m; \\ 0, & \text{otherwise;} \end{cases}$$

and

$$B_n(q) := \begin{cases} \sum_{k=0}^{m} \begin{bmatrix} 2m+1 \\ 2k \end{bmatrix}_q (-1)^{m+k} E_{2m-2k+1}(q) E_{2k}(q), & \text{if } n = 2m+1; \\ 0, & \text{otherwise.} \end{cases}$$

Equivalently,

$$\sum_{n\geqslant 0} A_{4n}(q) \frac{x^{4n}}{(q;q)_{4n}} = \sec_q(x)\mathrm{sech}_q(x) \quad \text{and} \quad \sum_{n\geqslant 0} B_{2n+1}(q) \frac{x^{2n+1}}{(q;q)_{2n+1}} = \sec_q(x)\tanh_q(x),$$

where $\mathrm{sech}_q(x) := \sec_q(ix)$ and $\tanh_q(x) := -i\tan_q(ix)$. The first few values of $A_{4n}(q)$ and $B_{2n+1}(q)$ are:

$$\begin{aligned} A_0(q) &= 1, \\ A_4(q) &= (1+q)(q^3+2q-1), \\ A_8(q) &= (1+q)(1+q^3)q^2(q^{18}+2q^{17}+7q^{16}+10q^{15}+18q^{14}+21q^{13}+29q^{12} \\ &\quad +30q^{11}+36q^{10}+31q^9+30q^8+23q^7+19q^6+11q^5+7q^4+2q^3-1) \end{aligned}$$

and

$$\begin{aligned} B_1(q) &= 1, \\ B_3(q) &= 1, \\ B_5(q) &= q^8+2q^7+4q^6+5q^5+4q^4+3q^3+2q^2, \\ B_7(q) &= q^{16}+2q^{15}+5q^{14}+8q^{13}+11q^{12}+15q^{11}+18q^{10}+18q^9+18q^8 \\ &\quad +16q^7+12q^6+9q^5+5q^4+2q^3+q^2, \\ B_9(q) &= q^{32}+4q^{31}+13q^{30}+31q^{29}+62q^{28}+110q^{27}+177q^{26}+264q^{25}+370q^{24} \\ &\quad +487q^{23}+607q^{22}+720q^{21}+815q^{20}+881q^{19}+911q^{18}+901q^{17} \\ &\quad +852q^{16}+770q^{15}+665q^{14}+547q^{13}+427q^{12}+314q^{11}+217q^{10} \\ &\quad +140q^9+82q^8+44q^7+20q^6+7q^5+2q^4. \end{aligned}$$

Based on empirical evidence, we propose the following conjecture for further investigation. Recall that a polynomial $F(x) = \sum_{k=m}^{M} f_k x^k$ is said to be *(weakly) spiral* if its coefficients satisfy the following inequalities:

$$f_{M-j} \leqslant f_{m+j} \leqslant f_{M-j-1}, \ \forall\ 0 \leqslant j \leqslant \lfloor (M-m-1)/2 \rfloor.$$

Conjecture 5.13. *For all $n \geqslant 0$,*

(1) $(-q;q^2)_n$ is a divisor of $A_{4n}(q)$;
(2) $B_{2n+1}(q) \in \mathbb{N}[q]$ is spiral, and in particular, it is unimodal.

In the following, we aim to present two combinatorial identities relating $A_{4n}(q)$ with $B_{2n+1}(q)$ and $E_{2n}(q)$.

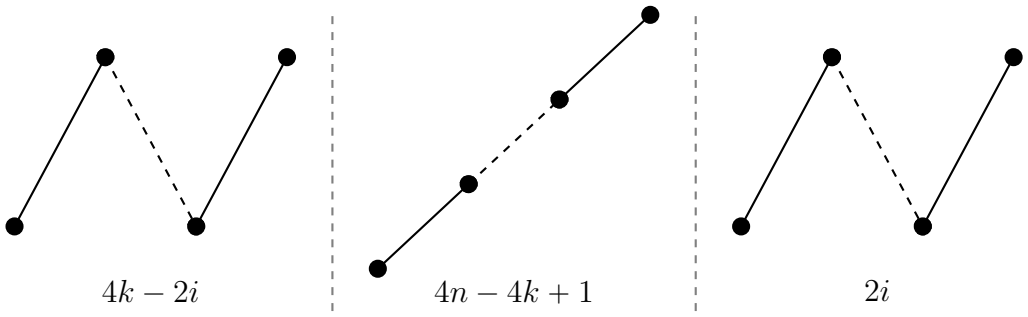


FIGURE 6. An illustration of triples in $\mathcal{B}(n,k,i)$.

Theorem 5.14. *For $n \geqslant 0$, we have*

$$\sum_{k=0}^{\lfloor n/2 \rfloor} \begin{bmatrix} 2n+1 \\ 4k \end{bmatrix}_q A_{4k}(q) = B_{2n+1}(q). \tag{5.17}$$

Proof. Note that (5.17) is equivalent to the following two identities:

$$\sum_{k=0}^{n}\sum_{i=0}^{2k} \begin{bmatrix} 4n+1 \\ 4k \end{bmatrix}_q \begin{bmatrix} 4k \\ 2i \end{bmatrix}_q (-1)^i E_{4k-2i}(q)E_{2i}(q) = \sum_{i=0}^{2n} \begin{bmatrix} 4n+1 \\ 2i \end{bmatrix}_q (-1)^i E_{4n-2i+1}(q)E_{2i}(q).$$

and

$$\sum_{k=0}^{n}\sum_{i=0}^{2k} \begin{bmatrix} 4n+3 \\ 4k \end{bmatrix}_q \begin{bmatrix} 4k \\ 2i \end{bmatrix}_q (-1)^i E_{4k-2i}(q)E_{2i}(q) = \sum_{i=0}^{2n+1} \begin{bmatrix} 4n+3 \\ 2i \end{bmatrix}_q (-1)^{i+1} E_{4n-2i+3}(q)E_{2i}(q).$$

Due to the similarity, we only prove combinatorially the first equality.

Let $\mathcal{B}(n,k,i)$ be the set of triples (a,b,c) of words satisfying

- $a = a_1a_2\ldots a_{4k-2i}$ and $c = c_1c_2\ldots c_{2i}$ are up-down sequences;
- $b = b_1b_2\ldots b_{4n-4k+1}$ is increasing;
- and the word concatenation $a\cdot b\cdot c$ is a permutation in $\mathfrak{S}_{4n+1}$.

See Fig. 6 for an illustration of such triples. Let $\mathcal{B}_n := \bigcup_{k,i} \mathcal{B}(n,k,i)$. By (2.3) and (2.4), we have

$$\sum_{(a,b,c)\in\mathcal{B}_n} (-1)^{|c|/2} q^{\mathsf{inv}(a\cdot b\cdot c)} = \sum_{k=0}^{n}\sum_{i=0}^{2k} \begin{bmatrix} 4n+1 \\ 4k \end{bmatrix}_q \begin{bmatrix} 4k \\ 2i \end{bmatrix}_q (-1)^i E_{4k-2i}(q)E_{2i}(q).$$

It remains to construct an involution η on $\mathcal{B}_n$ to prove that

$$\sum_{(a,b,c)\in\mathcal{B}_n} (-1)^{|c|/2} q^{\mathsf{inv}(a\cdot b\cdot c)} = \sum_{i=0}^{2n} \begin{bmatrix} 4n+1 \\ 2i \end{bmatrix}_q (-1)^i E_{4n-2i+1}(q)E_{2i}(q).$$

Given $(a,b,c) \in \mathcal{B}(n,k,i)$, define $\eta(a,b,c)$ to be the triple obtained as follows. For convenience when making comparison, we adopt the convention that $a_{4k-2i} := \infty$ when $a = \varnothing$, while $c_1 := 0$ when $c = \varnothing$.

- **Case 1:** $k \neq n$.
  - i) If a is non-empty with $a_{4k-2i} < b_1$, then move $a_{4k-2i-1}a_{4k-2i}$ from a to the beginning of b; otherwise, move b_1b_2 from b to the end of a.

ii) If c is non-empty with $c_1 > b_{4n-4k+1}$, then move c_1c_2 from c to the end of b; otherwise, move $b_{4n-4k}b_{4n-4k+1}$ from b to the beginning of c.

- **Case 2:** $k = n$ (i.e., $b = b_1$) and $a_{4k-2i} < b_1 < c_1$. Move $a_{4k-2i-1}a_{4k-2i}$ from a to the beginning of b and move c_1c_2 from c to the end of b.
- **Case 3:** $k = n$ and $a_{4k-2i} < b_1 > c_1$. In this case, we perform the following steps to get $\eta(a, b, c)$:
  (i) Move $a_{4k-2i-1}a_{4k-2i}$ from a to the beginning of b, yielding a' and b' with $|b'| = 3$.
  (ii) Move the last two entries of b' to the beginning of c, yielding b^* with $|b^*| = 1$ and c^*.
  (iii) Define $\eta(a, b, c) = (a', b^*, c^*)^{rc}$. Note that the image $\eta(a, b, c) \in \mathcal{B}(n, k, 2k-1-i)$ remains in Case 3.
- **Case 4:** $k = n$ and $a_{4k-2i} > b_1$. Define $\eta(a, b, c) = (a, b, c)$.

For example, let $\pi := (1534, 8, 6729) \in \mathcal{B}(2, 2, 2)$ be a triple covered by case 3. Then its image $\eta(\pi) = (183426, 7, 59) \in \mathcal{B}(2, 2, 1)$ is in case 3 as well.

One can verify that η is an inversion number preserving involution on $\mathcal{B}_n$ fixing elements in $\mathcal{B}_n^*$ and sign-reversing on $\mathcal{B}_n \backslash \mathcal{B}_n^*$, where $\mathcal{B}_n^*$ denotes the set of all triples $(a, b, c) \in \mathcal{B}_n$ satisfying $|b| = 1$ and the last letter of a is greater than b_1. Thus,

$$\begin{aligned}\sum_{(a,b,c)\in\mathcal{B}_n} (-1)^{|c|/2} q^{\mathsf{inv}(a\cdot b\cdot c)} &= \sum_{(a,b,c)\in\mathcal{B}_n^*} (-1)^{|c|/2} q^{\mathsf{inv}(a\cdot b\cdot c)} \\ &= \sum_{i=0}^{2n} \begin{bmatrix} 4n+1 \\ 2i \end{bmatrix}_q (-1)^i E_{4n-2i+1}(q) E_{2i}(q),\end{aligned}$$

which completes the proof of the first equality. □

Using essentially the same approach as in Theorem 5.14, we can prove combinatorially the following identity.

Theorem 5.15. *For $n \geqslant 0$, we have*

$$\sum_{k=0}^{\lfloor n/2 \rfloor} \begin{bmatrix} 2n \\ 4k \end{bmatrix}_q A_{4k}(q) = E_{2n}(q). \tag{5.18}$$

6. On combinatorics of Sun-like identities

6.1. **On Sun's Legendre-transformation identity.** The *n-th Legendre polynomial* $P_n(x)$ can be defined by

$$P_n(x) := \sum_{k=0}^{n} \binom{n}{k} \binom{n+k}{k} \left(\frac{x-1}{2}\right)^k.$$

It is known (see [39]) that the generating function for the Legendre polynomials is

$$\sum_{n\geqslant 0} P_n(x) t^n = (1 - 2xt + t^2)^{-\frac{1}{2}}. \tag{6.1}$$

The Legendre polynomials possess the following symmetry

$$P_n(-x) = (-1)^n P_n(x), \tag{6.2}$$

namely

$$\sum_{k=0}^{n}\binom{n}{k}\binom{n+k}{k}\left(\frac{-x-1}{2}\right)^k=(-1)^n\sum_{k=0}^{n}\binom{n}{k}\binom{n+k}{k}\left(\frac{x-1}{2}\right)^k.$$

Setting $\frac{x-1}{2}=y$ gives

$$\sum_{k=0}^{n}\binom{n}{k}\binom{n+k}{k}(-y-1)^k=(-1)^n\sum_{k=0}^{n}\binom{n}{k}\binom{n+k}{k}y^k,$$

which is equivalent to

$$\sum_{l=k}^{n}\binom{n}{l}\binom{n+l}{l}\binom{l}{k}(-1)^{n-l}=\binom{n}{k}\binom{n+k}{k} \tag{6.3}$$

for $k=0,1,\ldots,n$. This shows that the symmetry (6.2) of Legendre polynomials is indeed equivalent to the following identity proved by Sun [36].

Theorem 6.1 (Sun's Legendre-transformation identity)**.** *Let* $y_k=\sum_{l=0}^{k}\binom{k}{l}x_l$. *Then,*

$$\sum_{k=0}^{n}\binom{n}{k}\binom{n+k}{k}x_k=\sum_{k=0}^{n}\binom{n}{k}\binom{n+k}{k}(-1)^{n-k}y_k. \tag{6.4}$$

Sun [37] derived his secant-number identity (1.6) from the above theorem. As (6.3) and (6.4) are equivalent, we will provide two involutive proofs of (6.3), which constitutes the first step of our combinatorial proof of (1.6).

6.2. **An identity relating tangent numbers and secant numbers.** The classical Eulerian polynomials $A_n(t)$ can be defined as the descent polynomials

$$A_n(t):=\sum_{\pi\in\mathfrak{S}_n}t^{\mathsf{des}(\pi)+1},$$

where $\mathsf{des}(\pi):=|\{i\in[n-1]:\pi_i>\pi_{i+1}\}|$ is the *descent number* of π for each $\pi\in\mathfrak{S}_n$. For the sake of convenience, set $A_0(t)=1$. The *binomial-Eulerian polynomials* $\tilde{A}_n(t)$ were considered by Shareshian and Wachs [26]

$$\tilde{A}_n(t):=\sum_{m=0}^{n}\binom{n}{m}A_m(t). \tag{6.5}$$

The following identity relating Eulerian polynomials with the binomial-Eulerian polynomials is an immediate consequence of Theorem 6.1.

Theorem 6.2. *For* $n\geqslant 0$, *we have*

$$\sum_{k=0}^{n}\binom{n}{k}\binom{n+k}{k}A_k(t)=\sum_{k=0}^{n}\binom{n}{k}\binom{n+k}{k}(-1)^{n-k}\tilde{A}_k(t). \tag{6.6}$$

Setting $t=-1$ in (6.6) and then applying identity (3.3) and [23, Theorem 1.2] results in the following identity, which relates tangent numbers with secant numbers.

Corollary 6.3. *For* $n\geqslant 0$, *we have*

$$1+\sum_{k=0}^{\lfloor\frac{n}{2}\rfloor}\binom{n}{2k+1}\binom{n+2k+1}{2k+1}(-1)^{k+1}E_{2k+1}=\sum_{k=0}^{\lfloor\frac{n}{2}\rfloor}\binom{n}{2k}\binom{n+2k}{2k}(-1)^{n-k}E_{2k}. \tag{6.7}$$

There is an alternative approach to (6.7). By (2.7) and (2.8), we have

$$\sum_{k\geqslant 0}\binom{n}{2k}(-1)^k E_{2k}=\begin{cases}1, & \text{if } n=0;\\ 0, & \text{if } n>0 \text{ is even};\\ (-1)^m E_{2m+1}, & \text{if } n=2m+1.\end{cases}$$

In view of the above identity, setting $x_{2k}=(-1)^k E_{2k}$ and $x_{2k+1}=0$ in (6.4) provides another proof of (6.7).

6.3. **First involutive proof of (6.3).** The identity (6.3) can be recast as

$$\sum_{l=k}^{n}\binom{n}{n-l}\left(\!\binom{n+1}{l}\!\right)\binom{l}{k}(-1)^{n-l}=\binom{n}{k}\binom{n+k}{k}. \tag{6.8}$$

Consider the set $\mathcal{L}_{n,k}$ of all pairs (σ,w) of words, where σ is an increasing word over $[n]$ with distinct letters and w is a weakly increasing word over $[n+1]$ with exactly k of its letters receiving bars and satisfying $|\sigma|+|w|=n$. For instance, the pair $(358,\bar{2}2\bar{3}\bar{3}3\bar{6})$ is an element in $\mathcal{L}_{10,3}$. It is clear that the left-hand side of (6.8) can be interpreted as the following signed counting of $\mathcal{L}_{n,k}$:

$$\sum_{l=k}^{n}\binom{n}{n-l}\left(\!\binom{n+1}{l}\!\right)\binom{l}{k}(-1)^{n-l}=\sum_{(\sigma,w)\in\mathcal{L}_{n,k}}(-1)^{|\sigma|}.$$

Given $(\sigma,w)\in\mathcal{L}_{n,k}$, suppose that w_m (may not exist) is the smallest unbarred letter in w such that $w_{m-1}<w_m$ (according to their numerical value, and $w_0=0$ by convention) and $w_m\neq n+1$. For instance, if $(\sigma,w)=(358,\bar{2}2\bar{3}\bar{3}3\bar{6})$, then $w_m=w_3=3$. For the sake of convenience, we set $\sigma_1=\infty$ if σ is empty and $w_m=\infty$ if such w_m does not exist. Define the mapping $f:(\sigma,w)\mapsto(\sigma',w')$ as

(i) if $\sigma_1=w_m=\infty$, then set $(\sigma',w')=(\sigma,w)$;
(ii) if $\infty\neq\sigma_1\leqslant w_m$, then (σ',w') is obtained from (σ,w) by taking σ_1 from σ and inserting it into w as left as possible so that the resulting w' is weakly increasing;
(iii) if $\sigma_1>w_m$, then (σ',w') is obtained from (σ,w) by taking w_m from w and inserting it as the first letter of σ.

It is plain to see that f is an involution on $\mathcal{L}_{n,k}$ which fixes elements in case (i) and reverses the signs of the other elements. It remains to show that the elements in case (i) has cardinality $\binom{n}{k}\binom{n+k}{k}$.

The elements (σ,w) in $\mathcal{L}_{n,k}$ falling into case (i) are those with $\sigma=\varnothing$ and w being a weakly increasing word, where the first appearance of each value (except $n+1$) is barred. For instance, the pair $(\varnothing,\bar{2}2\bar{3}3\bar{3}\bar{5}99)\in\mathcal{L}_{8,4}$ is in case (i). Such elements can be constructed in the following steps:

- choose k (repetitions allowed) barred letters from $[n+1]$, mark them with bars, and arrange them in weakly increasing order;
- put a box immediately after each barred letters and label it using the value of that letter and put an extra box labeled $n+1$ immediately before the first barred $n+1$ (if there is no barred $n+1$, then just put the extra box at the end);
- choose $n-k$ boxes (repetitions allowed) from the above $k+1$ boxes;

- finally, insert c_i copies of l_i into the i-th box if the i-th box, whose label is l_i, was chosen c_i times in the last step for $i = 1, 2, \ldots, k+1$.

For example, to form $(\varnothing, \bar{2}2\bar{3}3\bar{3}\bar{5}99)$, we first choose $\bar{2}\bar{3}\bar{3}\bar{5}$, which becomes

$$\bar{2}\square_2\bar{3}\square_3\bar{3}\square_3\bar{5}\square_5\square_9$$

after putting boxes. Now if we choose $c_1 = 1, c_2 = 1, c_3 = 0, c_4 = 0$ and $c_5 = 2$, then we obtain the word $\bar{2}2\bar{3}3\bar{3}\bar{5}99$ after inserting letters. From the above construction, we see that the number of fixed elements of f in $\mathcal{L}_{n,k}$ is exactly

$$\left(\!\!\binom{n+1}{k}\!\!\right)\left(\!\!\binom{k+1}{n-k}\!\!\right) = \binom{n+k}{k}\binom{n}{n-k},$$

as desired.

6.4. **Second involutive proof of** (6.3)**.** The sum $\sum_{k=0}^{n}\binom{n}{k}\binom{n+k}{k}$ is registered in the OEIS [27] as the sequence A001850 and is called the n-th *central Delannoy number*. Our interpretation of it via the signed counting of the pairs $\bigcup_{0\leqslant k\leqslant n}\mathcal{L}_{n,k}$, as shown in (6.8), appears to be new. Alternatively, we can interpret this number as follows; see [22] for an equivalent interpretation in terms of *free Schröder path*. Recall that for any nonnegative integer n and any integer m, a path in the integer lattice $\mathbb{Z}^2$ from $(0,0)$ to (n,m) is called a *Delannoy path* of length n, if the allowed steps are north $\mathrm{N}: (x,y)\to(x,y+1)$; east $\mathrm{E}: (x,y)\to(x+1,y)$; or northeast $\mathrm{L}: (x,y)\to(x+1,y+1)$. We often use a word consisting of letters $\mathrm{N}, \mathrm{E}, \mathrm{L}$ to represent a Delannoy path. A Delannoy path ending at (n,n) without any northeast steps is called a *free Dyck path* of length n. The *height* (with respect to the line $y=x$) of a step $(x,y)\to(x',y')$ is taken to be $y-x$. A peak (resp. valley) in a path refers to a consecutive pair of steps NE (resp. EN). The height of a peak or a valley is taken to be the height of its first step.

Definition 6.4. *For $0\leqslant k\leqslant l\leqslant n$, we denote by $\mathcal{D}_{n,l,k}$ the set of Delannoy paths ending at (n,n), having l north steps, k of which are colored (say, in blue). A peak (resp. valley) is called colored if the north step contained in it is colored.*

Proposition 6.5. *For $0\leqslant k\leqslant l\leqslant n$, we have*

$$|\mathcal{D}_{n,l,k}| = \binom{n}{l}\binom{n+l}{l}\binom{l}{k}. \tag{6.9}$$

Proof. Every colored Delannoy path $p\in\mathcal{D}_{n,l,k}$ can be constructed in three steps as follows.

- Construct a free Dyck path of length l. There are $\binom{2l}{l}$ possibilities.
- Choose k north steps to be colored blue. Denote this colored free Dyck path as $\tilde{p} = w_1w_2\cdots w_{2l}$. There are $\binom{l}{k}$ possibilities.
- Turning $\tilde{p}$ into the desired Delannoy path p by inserting $n-l$ northeast steps into $2l+1$ slots (i.e., the place before w_1, the place inbetween w_i and w_{i+1} for $1\leqslant i\leqslant 2l-1$, and the place after w_{2l}). Note that the number of possibilities for this final step is given by $\left(\!\!\binom{2l+1}{n-l}\!\!\right) = \binom{n+l}{n-l}$.

To finish the proof, one can verify the following simple equality:

$$\binom{2l}{l}\binom{l}{k}\binom{n+l}{n-l} = \binom{n}{l}\binom{n+l}{l}\binom{l}{k}.$$

□

We define an involution ϱ on the union $\bigcup_{k\leqslant l\leqslant n}\mathcal{D}_{n,l,k}$ as follows. We scan the given path $p = w_1\cdots w_{n+l}$ step by step, find the smallest index i such that one of the following conditions is satisfied.

(1) w_iw_{i+1} = NE is an uncolored peak at height $h \geqslant 0$.
(2) w_iw_{i+1} = EN is an uncolored valley at height $h < 0$.
(3) w_i = L is a northeast step at height h.

In case (1), we replace the peak w_iw_{i+1} with a northeast step L at height $h \geqslant 0$, and take this new path $\hat{p} \in \mathcal{D}_{n,l-1,k}$ to be the image $\varrho(p)$; in case (2), we replace the valley w_iw_{i+1} with a northeast step L at height $h < 0$, and take this new path $\hat{p} \in \mathcal{D}_{n,l-1,k}$ to be the image $\varrho(p)$; in case (3), we replace w_i with a peak NE at height h when $h \geqslant 0$, or we replace w_i with a valley EN at height h when $h < 0$, taking the new path $\hat{p} \in \mathcal{D}_{n,l+1,k}$ thus obtained to be the image $\varrho(p)$. When none of cases (1), (2), (3) happens, we simply set $\varrho(p) = p$. Those paths that are fixed by ϱ can be characterized as

$$\mathcal{D}^*_{n,k} := \{p \in \mathcal{D}_{n,n,k} : \text{all peaks of height } h \geqslant 0 \text{ and valleys of height } h < 0 \text{ are colored}\}.$$

The construction of ϱ shows that it is indeed an involution swapping the paths in cases (1) and (2) with those in case (3). It keeps the length of the path n, as well as the number of colored steps k, while it changes the number of northeast steps by one. Consequently, if we weight each Delannoy path $p \in \mathcal{D}_{n,l,k}$ by $(-1)^{n-l}$, then ϱ explains the following relation combinatorially.

Lemma 6.6. *For $0 \leqslant k \leqslant l \leqslant n$, we have*

$$\sum_{l=k}^{n}\binom{n}{l}\binom{n+l}{l}\binom{l}{k}(-1)^{n-l} = \sum_{k\leqslant l\leqslant n}\sum_{p\in\mathcal{D}_{n,l,k}}(-1)^{n-l} = |\mathcal{D}^*_{n,k}|. \tag{6.10}$$

Now we see that to establish (6.3), it remains to show the following lemma.

Lemma 6.7. *For $0 \leqslant k \leqslant l \leqslant n$, we have*

$$|\mathcal{D}^*_{n,k}| = \binom{n}{k}\binom{n+k}{k}. \tag{6.11}$$

Our approach is to first construct a bijection between $\mathcal{D}^*_{n,k}$ and $\mathcal{D}_{n,k,0}$ and then apply (6.9) to get the desired formula shown in (6.11).

Recall that a Dyck path is called *raised* if it does not touch the line $y = x$ until its final step. In other words, a Dyck path p is raised if and only if it can be decomposed as p = Np'E (or p = Ep'N), with p' being a Dyck path itself. Clearly each free Dyck path (colored or not) can be uniquely decomposed into *components*, such that each component is itself a raised Dyck path which stays either strictly above the line $y = x$ or strictly below the line $y = x$. The notion of *raised Delannoy path* can be similarly introduced.

First proof of Lemma 6.7. We are going to construct a bijection $\theta : \mathcal{D}^*_{n,k} \to \mathcal{D}_{n,k,0}$ inductively. Firstly the empty path is mapped to the empty path. Next for $n = 1$ and 2, the correspondences are illustrated in Fig. 7.

Given any path $p \in \mathcal{D}^*_{n,k}$ with the decomposition $p = C_1 \cdots C_m$, where each C_i, $1 \leqslant i \leqslant m$, is a raised and peak-colored (resp., valley-colored) Dyck path above $y = x$ (resp., below $y = x$). If $m \geqslant 2$, then we let

$$\theta(p) := \theta(C_1)\theta(C_2)\cdots\theta(C_m)$$

and use induction since now each C_i is shorter than p. It remains to construct $\theta(p)$ for the case of $m = 1$, i.e., p itself is raised. The mapping for the case of p with length 1 is shown in the first line of Fig. 7. We will only explain the case when p has length $n \geqslant 2$ and is above the line $y = x$, since the case when p is below the line $y = x$ is just a mirror reflection[2] with respect to the line $y = x$. Suppose $p = \mathrm{N}_1 p' \mathrm{E}_1$ and $p' = (\mathrm{N}_2 p'' \mathrm{E}_2) p'''$, where $\mathrm{N}_2 p'' \mathrm{E}_2$ is the first component of p' and both p'' and p''' could be empty. There are two subcases.

(1) The step N_1 is colored, then we set $\theta(p) := \mathrm{N}\theta(p')\mathrm{E}$.
(2) The step N_1 is uncolored. We denote $q := \theta(\mathrm{N}_2 p'' \mathrm{E}_2)$ which is a raised Delannoy path (by induction). Thus we can write $q = \mathrm{N}q'\mathrm{E}$ with q' being a Delannoy path staying weakly above $y = x$. Now we set $\theta(p) := \mathrm{N}q'\mathrm{L}\theta(p''')\mathrm{E}$.

To construct the inverse of θ, it is again sufficient to explain how to recover the preimage $\theta^{-1}(\tilde{p})$ when the given path $\tilde{p} \in \mathcal{D}_{n,k,0}$ is a raised Delannoy path above the line $y = x$. Note that the image paths $\theta(p)$ obtained from case (1) above do not have a northeast step at height 1, while those image paths from case (2) must have at least one northeast step at height 1. This key observation leads to the following construction of θ^{-1} in two subcases.

(1) $\tilde{p}$ does not have any height 1 northeast steps, then we write $\tilde{p} = \mathrm{N}\tilde{p}'\mathrm{E}$ and set $\theta^{-1}(\tilde{p}) := \mathrm{N}\theta^{-1}(\tilde{p}')\mathrm{E}$ with the first step N being colored.
(2) $\tilde{p}$ does have a height 1 northeast step. Find the rightmost such step L_0 and decompose as $\tilde{p} = \mathrm{N}\tilde{p}'\mathrm{L}_0\tilde{p}''\mathrm{E}$, where $\tilde{p}''$ is raised (or empty) due to our choice of L_0. We set $\theta^{-1}(\tilde{p}) := \mathrm{N}_1\theta^{-1}(\mathrm{N}_2\tilde{p}'\mathrm{E}_2)\theta^{-1}(\tilde{p}'')\mathrm{E}_1$, where the first step N_1 is uncolored.

We trust the reader to verify that the above construction indeed gives us the inverse of θ. □

Now in view of Lemma 6.7, plugging in $x = 3$ in (6.1) gives us the generating function for the sum $\sum_{0\leqslant k\leqslant n} |\mathcal{D}^*_{n,k}|$:

$$\sum_{n\geqslant 0}\left(\sum_{0\leqslant k\leqslant n} |\mathcal{D}^*_{n,k}|\right) z^n = \sum_{n\geqslant 0}\sum_{0\leqslant k\leqslant n}\binom{n}{k}\binom{n+k}{k} z^n = \sum_{n\geqslant 0} P_n(3) z^n = \frac{1}{\sqrt{1-6z+z^2}}. \tag{6.12}$$

We include here an algebraic proof of (6.12) without using Lemma 6.7. To that end, we require the following interpretation of the *Narayana polynomial*. Let $\mathcal{D}_n$ denote the set of Dyck paths of length n (i.e., those paths in $\mathcal{D}_{n,n,0}$ which never go below the line $y = x$).

Proposition 6.8. *(i) The distribution of the number of peaks (denoted as* $\mathrm{pk}(p)$ *for a given path* $p \in \mathcal{D}_n$*) over* $\mathcal{D}_n$ *is given by the Narayana polynomial:*

$$N_n(t) := \sum_{k=1}^{n} \frac{1}{k}\binom{n}{k-1}\binom{n-1}{k-1} t^k = \sum_{p\in\mathcal{D}_n} t^{\mathrm{pk}(p)}.$$

[2]To be precise, the mirror reflection ρ is a mapping on words in $\{N, E\}^*$ that switches N and E. For example, $\rho(NNNEENEE) = EEENNENN$. Now for any raised p below $y = x$ with length greater than one, we set $\theta(p) = \rho(\theta(\rho(p)))$.

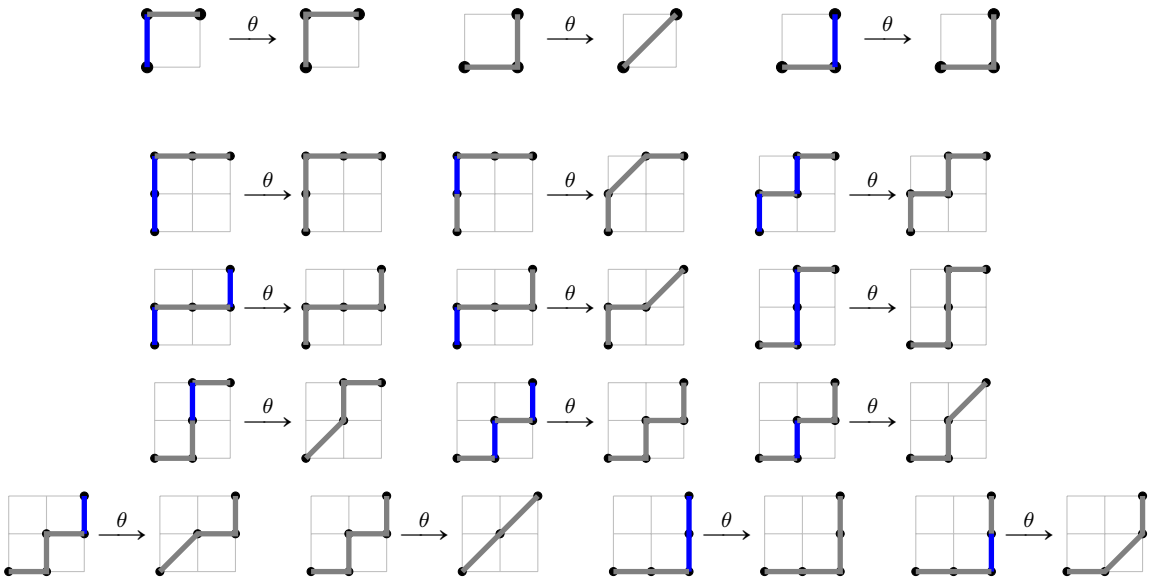


FIGURE 7. The correspondences under θ for $n = 1, 2$

(ii) Let $\mathcal{D}_n^\dagger$ denote the set of paths in $\mathcal{D}_n^* := \bigcup_{0\leqslant k\leqslant n} \mathcal{D}_{n,k}^*$ which never go below the line $y = x$ and set $D_n^\dagger := |\mathcal{D}_n^\dagger|$. We have the following explicit expression for the generating function:

$$\sum_{n\geqslant 0} D_n^\dagger z^n = \frac{1 + z - \sqrt{1 - 6z + z^2}}{4z}. \tag{6.13}$$

Proof. See [28, Chapter 2.4.2] for a proof of (i). For (ii), it suffices to note that for a path in $\mathcal{D}_n^\dagger$, every north step that is not included in a peak can be either colored or uncolored, hence we have the following relation for all $n \geqslant 0$:

$$D_n^\dagger = 2^n N_n(2^{-1}) = \sum_{k=1}^{n} \frac{1}{k}\binom{n}{k-1}\binom{n-1}{k-1} 2^{n-k}.$$

Moreover, recall the generating function of Narayana polynomials [28, Eq. (2.6)]:

$$\sum_{n\geqslant 0} N_n(t) z^n = \frac{1 + z(1-t) - \sqrt{1 - 2z(1+t) + z^2(1-t)^2}}{2z}.$$

Plugging in $z \to 2z$ and $t \to 1/2$ gives us (6.13) after simplification. □

Second proof of (6.12). The idea is to establish a recurrence relation for the cardinality $D_n^* := |\mathcal{D}_n^*|$, and derive its generating function basing on that recurrence relation. Given a path $p \in \mathcal{D}_n^*$, we take its "first return decomposition", i.e., we write $p = p'p''$, where p' is a raised and colored Dyck path above $y = x$ or below $y = x$. Suppose the length of p' is m, $1 \leqslant m \leqslant n$. If $m = 1$ there are three cases, namely, $p' =$ NE colored, $p' =$ EN uncolored, and $p' =$ EN colored. If $m \geqslant 2$, we can further decompose as $p' = \mathrm{N}p'''\mathrm{E}$ if p' is above $y = x$, or we have $p' = \mathrm{E}p'''\mathrm{N}$ if p' is below $y = x$. In both cases the step N could be either colored or uncolored, so there are four possibilities in total. Moreover, for the

"above" case (resp., "below" case), the subpath p''' (resp., the mirror reflection of p''' with respect to $y = x$) belongs to $\mathcal{D}^{\dagger}_{m-1}$. The above analysis leads to the following recurrence for $n \geqslant 1$:

$$\begin{aligned} D^*_n &= 3D^*_{n-1} + 4 \sum_{2 \leqslant m \leqslant n} D^{\dagger}_{m-1} D^*_{n-m} \\ &= -D^*_{n-1} + 4 \sum_{1 \leqslant m \leqslant n} D^{\dagger}_{m-1} D^*_{n-m}. \end{aligned}$$

Writing $D^*(z) = \sum_{n \geqslant 0} D^*_n z^n$ and $D^{\dagger}(z) = \sum_{n \geqslant 0} D^{\dagger}_n z^n$, the above recurrence can be turned into the following functional equation.

$$D^*(z) - 1 = -zD^*(z) + 4zD^{\dagger}(z)D^*(z).$$

Plugging in (6.13) and solving for $D^*(z)$, we get (6.12) as desired. □

6.5. **Two combinatorial proofs of (1.6).** The following identity is a consequence of the generating function (6.1) for the Legendre polynomials. We provide an involutive proof of it, as the same idea will be used in some of its modifications.

Lemma 6.9. *For $m \geqslant 0$, we have*

$$\sum_{k=0}^{m} \binom{m}{k} \binom{m+k}{k} (-1)^k 2^{m-k} = \begin{cases} 0, & \text{if } m \text{ is odd;} \\ (-1)^n \binom{2n}{n}, & \text{if } m = 2n. \end{cases} \tag{6.14}$$

Proof. Rewrite the left-hand side of (6.14) as

$$\sum_{k=0}^{m} \binom{m}{m-k} \left(\!\!\binom{m+1}{k}\!\!\right) (-1)^k 2^{m-k}. \tag{6.15}$$

It can be interpreted as the signed counting

$$\sum_{(S,T) \in \mathcal{L}_m} (-1)^{|T|},$$

where $\mathcal{L}_m$ is the set of pairs (S, T) with S a subset, whose elements are colored red or black, of $[m]$ and T is a multisubset of $[m+1]$ such that $|S| + |T| = m$. For instance, $(\{\mathbf{2}, 4, 5, \mathbf{7}\}, \{2, 2, 8\})$ is an element in $\mathcal{L}_7$. Given $(S, T) \in \mathcal{L}_m$, let s be the smallest black element in S and let t be the smallest element in T such that $t \notin S \cup \{m+1\}$. Set s or t to be ∞ if such a smallest element does not exist. Consider the mapping $f : (S, T) \mapsto (S', T')$ in the following three cases:

(i) if $s = t = \infty$, then $(S', T') = (S, T)$;
(ii) if $s < t$, then (S', T') is obtained from (S, T) by moving s from S to T;
(iii) if $s > t$, then (S', T') is obtained from (S, T) by moving t from T to S.

For instance, if $(S, T) = (\{\mathbf{2}, 4, 5, \mathbf{7}\}, \{2, 2, 4, 8\})$, then $f(S, T) = (\{\mathbf{2}, 5, \mathbf{7}\}, \{2, 2, 4, 4, 8\})$. It is clear that f is an involution on $\mathcal{L}_m$ fixing pairs in case (i) while reversing the signs for pairs in the other cases. The pairs in case (i) above are those $(S, T) \in \mathcal{L}_m$ such that S is a subset with all elements red and all elements of T are either in S or equal to $m+1$. Thus,

$$\sum_{(S,T) \in \mathcal{L}_m} (-1)^{|T|} = \sum_{k=0}^{m} \binom{m}{m-k} \left(\!\!\binom{m-k+1}{k}\!\!\right) (-1)^k = \sum_{k=0}^{m} \binom{m}{m-k} \binom{m}{k} (-1)^k.$$

It remains to show that

$$\sum_{k=0}^{m}\binom{m}{m-k}\binom{m}{k}(-1)^k=\begin{cases}0, & \text{if } m \text{ is odd;}\\ (-1)^n\binom{2n}{n}, & \text{if } m=2n.\end{cases} \tag{6.16}$$

Interpret the left-hand side of (6.16) as

$$\sum_{k=0}^{m}\binom{m}{m-k}\binom{m}{k}(-1)^k=\sum_{(S,T)\in\mathcal{J}_m}(-1)^{|T|},$$

where $\mathcal{J}_m$ is the set of pairs (S,T) with $S,T\subseteq[m]$ and $|S|+|T|=m$. Now consider the involution $g:(S,T)\mapsto(S',T')$, where (S',T') is obtained from (S,T) by toggling between S and T the minimum element in the *symmetric difference*[3] $S\Delta T$. If m is odd, then $S\neq T$ so $S\Delta T\neq\varnothing$, hence g is a sign-reversing involution on $\mathcal{J}_m$; if $m=2n$ is even, g fixes the pairs (S,S) with S an n-element subset of $[2n]$ and reverses the signs of the other pairs. This proves (6.16). □

First combinatorial proof of (1.6). By (6.4), we have

$$\sum_{k=0}^{2n}\binom{2n}{k}\binom{2n+k}{k}x_k=\sum_{k=0}^{2n}\binom{2n}{k}\binom{2n+k}{k}(-1)^k y_k. \tag{6.17}$$

Setting $x_{2k}=2^{2n-2k}(-1)^kE_{2k}$ and $x_{2k+1}=0$ for $k\geqslant 0$ in (6.17) yields

$$\sum_{k=0}^{n}\binom{2n}{2k}\binom{2n+2k}{2k}2^{2n-2k}(-1)^kE_{2k}=\sum_{k=0}^{2n}\binom{2n}{k}\binom{2n+k}{k}(-1)^k y_k, \tag{6.18}$$

where

$$y_{2k}=\sum_{l=0}^{k}\binom{2k}{2l}2^{2n-2l}(-1)^lE_{2l}=2^{2n-2k}(2+(-1)^{k-1}E_{2k})\quad\text{by (3.13)} \tag{6.19}$$

and

$$y_{2k+1}=\sum_{l=0}^{k}\binom{2k+1}{2l}2^{2n-2l}(-1)^lE_{2l}=2^{2n-2k}\quad\text{by (3.12).} \tag{6.20}$$

Combining (6.18), (6.19) and (6.20) gives

$$\begin{aligned}&\sum_{k=0}^{n}\binom{2n}{2k}\binom{2n+2k}{2k}2^{2n-2k}(-1)^kE_{2k}\\ &=\sum_{k=0}^{n}\binom{2n}{2k}\binom{2n+2k}{2k}2^{2n-2k}-\sum_{k=0}^{n-1}\binom{2n}{2k+1}\binom{2n+2k+1}{2k+1}2^{2n-2k-1}\\ &=\sum_{k=0}^{2n}\binom{2n}{k}\binom{2n+k}{k}(-1)^k2^{2n-k}\\ &=(-1)^n\binom{2n}{n},\end{aligned}$$

[3] $S\Delta T:=(S\backslash T)\bigcup(T\backslash S)$.

where the last equality follows from Lemma 6.9. □

It is our attempt to derive a q-analog of (1.6) that leads to an alternative proof of this identity. Our second approach to (1.6) avoids the use of (6.4), but it requires the following combinatorial identity.

Lemma 6.10. *The following identity holds for any $m \geqslant j \geqslant 0$:*

$$\sum_{k=2j+1}^{2m} \binom{2m-2j-1}{2m-k} \binom{2m+k}{k} (-1)^k 2^{2m-k} = 0. \tag{6.21}$$

Proof. The proof is similar to that of Lemma 6.9. Rewrite the left-hand side of (6.21) as

$$\sum_{k=2j+1}^{2m} \binom{2m-2j-1}{2m-k} \left(\!\binom{2m+1}{k}\!\right) (-1)^k 2^{2m-k}. \tag{6.22}$$

It can be interpreted as the signed counting

$$\sum_{(S,T)\in\mathcal{F}_{m,j}} (-1)^{|T|},$$

where $\mathcal{F}_{m,j}$ is the set of pairs (S,T) with S a subset, whose elements are colored red or black, of $[2m-2j-1]$ and T is a multisubset of $[2m+1]$ such that $|S|+|T|=2m$. Given $(S,T)\in\mathcal{F}_{m,j}$, let s be the smallest black element in S and let t be the smallest element in T such that $t\notin S$ and $t<2m-2j$. Set s or t to be ∞ if such a smallest element does not exist. Consider the mapping $g:(S,T)\mapsto(S',T')$ in the following three cases:

(i) if $s=t=\infty$, then $(S',T')=(S,T)$;
(ii) if $s<t$, then (S',T') is obtained from (S,T) by moving s from S to T;
(iii) if $s>t$, then (S',T') is obtained from (S,T) by moving t from T to S.

It is clear that g is an involution on $\mathcal{F}_{m,j}$ fixing pairs in case (i) while reversing the signs for pairs in the other cases. The pairs in case (i) above are those $(S,T)\in\mathcal{F}_{m,j}$ such that S is a subset will all elements red and all elements of T are either in S or greater than $2m-2j-1$. Thus,

$$\begin{aligned}\sum_{(S,T)\in\mathcal{F}_{m,j}} (-1)^{|T|} &= \sum_{k=2j+1}^{2m} \binom{2m-2j-1}{2m-k} \left(\!\binom{2m-k+2j+2}{k}\!\right) (-1)^k \\ &= \sum_{k=2j+1}^{2m} \binom{2m-2j-1}{2m-k} \binom{2m+2j+1}{k} (-1)^k \\ &= 0,\end{aligned}$$

where the last equality follows from the observation that $a_{m,j}(2j+1+i) = -a_{m,j}(2m-i)$ for $0\leqslant i\leqslant m-j-1$ with

$$a_{m,j}(k) := \binom{2m-2j-1}{2m-k} \binom{2m+2j+1}{k} (-1)^k.$$

This completes the proof of the lemma. □

We can now provide a second proof of (1.6).

Second combinatorial proof of (1.6). Setting $q=1$ in (2.9) and (2.10) yields

$$T_n := \sum_{k\geqslant 0}\binom{n}{2k+1}(-1)^k E_{2k+1} = \begin{cases} 1, & \text{if } n \text{ is odd;}\\ 1+(-1)^{m+1}E_{2m}, & \text{if } n=2m.\end{cases} \tag{6.23}$$

On the one hand, we see that the following double summation vanishes:

$$\begin{aligned}&\sum_{k=0}^{2n}\binom{2n}{k}\binom{2n+k}{k}(-1)^k 2^{2n-k}\sum_{j\geqslant 0}\binom{k}{2j+1}(-1)^j E_{2j+1}\\ &=\sum_{j\geqslant 0}\binom{2n}{2j+1}(-1)^j E_{2j+1}\sum_{k=2j+1}^{2n}\binom{2n+k}{k}\binom{2n-2j-1}{2n-k}(-1)^k 2^{2n-k}=0,\end{aligned}$$

where we have applied (6.21) for the inner summation. On the other hand, the inner sum in the original expression is just T_k, so we can apply (6.23) to deduce that

$$\begin{aligned}0&=\sum_{k=0}^{2n}\binom{2n}{k}\binom{2n+k}{k}(-1)^k 2^{2n-k}T_k\\ &=\sum_{k=0}^{n}\binom{2n}{2k}\binom{2n+2k}{2k}2^{2n-2k}(1+(-1)^{k+1}E_{2k})-\sum_{k=0}^{n-1}\binom{2n}{2k+1}\binom{2n+2k+1}{2k+1}2^{2n-2k-1}.\end{aligned}$$

Or equivalently,

$$\begin{aligned}\sum_{k=0}^{n}\binom{2n}{2k}\binom{2n+2k}{2k}2^{2n-2k}(-1)^k E_{2k}&=\sum_{k=0}^{2n}\binom{2n}{k}\binom{2n+k}{k}2^{2n-k}(-1)^k\\ &=(-1)^n\binom{2n}{n},\end{aligned}$$

as desired. Here we have applied Lemma 6.9 for the last equality. □

6.6. **Some Sun-like identities for Euler numbers.** Our first identity is a tangent-number companion to Sun's secant-number identity (1.6).

Theorem 6.11. *For $n\geqslant 0$, we have*

$$\sum_{k=0}^{n}\binom{2n+1}{2k+1}\binom{2n+2k+2}{2k+1}2^{2n-2k}(-1)^k E_{2k+1}=2^{2n+1}. \tag{6.24}$$

Proof. By (6.4), we have

$$\sum_{k=0}^{2n+1}\binom{2n+1}{k}\binom{2n+1+k}{k}x_k=\sum_{k=0}^{2n+1}\binom{2n+1}{k}\binom{2n+1+k}{k}(-1)^{k-1}y_k. \tag{6.25}$$

Setting $x_{2k+1}=2^{2n-2k}(-1)^k E_{2k+1}$ and $x_{2k}=0$ for $k\geqslant 0$ in (6.25) yields

$$\sum_{k=0}^{n}\binom{2n+1}{2k+1}\binom{2n+2k+2}{2k+1}2^{2n-2k}(-1)^k E_{2k+1}=\sum_{k=0}^{2n+1}\binom{2n+1}{k}\binom{2n+1+k}{k}(-1)^{k-1}y_k,$$

where $y_0 = 0$,

$$y_{2k} = \sum_{l=0}^{k-1} \binom{2k}{2l+1} 2^{2n-2l}(-1)^l E_{2l+1} = 2^{2n+1} \quad \text{by (3.10)}$$

for $k \geqslant 1$, and

$$y_{2k+1} = \sum_{l=0}^{k} \binom{2k+1}{2l+1} 2^{2n-2l}(-1)^l E_{2l+1} = 2^{2n+1} + (-1)^{k-1} 2^{2n-2k} E_{2k+1} \quad \text{by (3.11)}$$

for $k \geqslant 0$. Combining the last three identities gives

$$\begin{aligned} &\sum_{k=0}^{n} \binom{2n+1}{2k+1} \binom{2n+2k+2}{2k+1} 2^{2n-2k}(-1)^k E_{2k+1} \\ &= 2^{2n} + 2^{2n} \sum_{k=0}^{2n+1} \binom{2n+1}{k} \binom{2n+1+k}{k} (-1)^{k-1} \\ &= 2^{2n+1}, \end{aligned}$$

where the last equality follows from the generating function (6.1) for the Legendre polynomials. This proves (6.24). □

Our next identity is analogous to Theorem 6.11.

Theorem 6.12. *For $n \geqslant 1$, we have*

$$\sum_{k=0}^{n} \binom{2n+1}{2k+1} \binom{2n+2k}{2k} 2^{2n-2k}(-1)^k E_{2k+1} = 0. \tag{6.26}$$

To prove Theorem 6.12, we require the following seemingly generalization of Theorem 6.1.

Theorem 6.13. *Fix an integer $d \geqslant 0$. Let $y_k = \sum_{l=0}^{k} \binom{k+d}{l+d} x_l$. Then,*

$$\sum_{k=0}^{n} \binom{n+d}{k+d} \binom{n+k}{k} x_k = \sum_{k=0}^{n} \binom{n+d}{k+d} \binom{n+k}{k} (-1)^{n-k} y_k. \tag{6.27}$$

Proof. We will deduce this generalization from Theorem 6.1. Setting $x_k = \frac{\tilde{x}_k}{\prod_{i=1}^{d}(k+i)}$ and $y_k = \frac{\tilde{y}_k}{\prod_{i=1}^{d}(k+i)}$ in (6.4) and multiplying both sides by $\prod_{i=1}^{d}(n+i)$ yields

$$\sum_{k=0}^{n} \binom{n+d}{k+d} \binom{n+k}{k} \tilde{x}_k = \sum_{k=0}^{n} \binom{n+d}{k+d} \binom{n+k}{k} (-1)^{n-k} \tilde{y}_k,$$

where

$$\tilde{y}_k = y_k \prod_{i=1}^{d}(k+i) = \prod_{i=1}^{d}(k+i) \sum_{l=0}^{k} \binom{k}{l} x_l = \sum_{l=0}^{k} \binom{k+d}{l+d} \tilde{x}_l.$$

This completes the proof of (6.27). □

We also need the following simple identity.

Lemma 6.14. *For $n \geqslant 1$, we have*

$$\sum_{k=0}^{n} \binom{n+1}{k+1}\binom{n+k}{k}(-1)^k = 0.$$

Proof. Let $\mathcal{P}_n$ be the set of all pairs (S,T) where S is a subset of $[n+1]$ and T is a multisubset of $[n+1]$ with $|S|+|T|=n$. For instance, $(\{1,2,4\},\{2,3,3,5\}) \in \mathcal{P}_7$. Then

$$\sum_{(S,T)\in\mathcal{P}_n} (-1)^{|T|} = \sum_{k=0}^{n} \binom{n+1}{n-k}\binom{n+k}{k}(-1)^k.$$

It remains to construct a sign-reversing involution on $\mathcal{P}_n$. Indeed, for a given pair (S,T), let us denote $s := \min(S)$ and $t := \min(T)$, and make the convention that $\min(\varnothing) = \infty$. We construct a new pair (S',T') according to the following two cases:

- If $s \leqslant t$, then set $S' = S\backslash\{s\}$ and $T' = T \cup \{s\}$.
- If $s > t$, then set $S' = S \cup \{t\}$ and $T' = T\backslash\{t\}$.

Note that since $|S|+|T| = n \geqslant 1$, they cannot both be empty, hence the mapping $(S,T) \mapsto (S',T')$ is a sign-reversing involution over $\mathcal{P}_n$ that explains the cancellation. □

We are ready for the proof of Theorem 6.12.

Proof of Theorem 6.12. By (6.27) with $d=1$, we have

$$\sum_{k=0}^{2n} \binom{2n+1}{k+1}\binom{2n+k}{k}x_k = \sum_{k=0}^{2n} \binom{2n+1}{k+1}\binom{2n+k}{k}(-1)^k y_k, \tag{6.28}$$

where $y_k = \sum_{l=0}^{k}\binom{k+1}{l+1}x_l$. Setting $x_{2k} = 2^{2n-2k}(-1)^k E_{2k+1}$ and $x_{2k+1} = 0$ for $k \geqslant 0$ in (6.28) yields

$$\sum_{k=0}^{n} \binom{2n+1}{2k+1}\binom{2n+2k}{2k}2^{2n-2k}(-1)^k E_{2k+1} = \sum_{k=0}^{2n} \binom{2n+1}{k+1}\binom{2n+k}{k}(-1)^k y_k,$$

where for $k \geqslant 0$

$$y_{2k} = \sum_{l=0}^{k} \binom{2k+1}{2l+1}2^{2n-2l}(-1)^l E_{2l+1} = 2^{2n+1} + (-1)^{k-1}2^{2n-2k}E_{2k+1} \quad \text{by (3.11)}$$

and for $k \geqslant 0$

$$y_{2k+1} = \sum_{l=0}^{k} \binom{2k+2}{2l+1}2^{2n-2l}(-1)^l E_{2l+1} = 2^{2n+1} \quad \text{by (3.10)}.$$

Combining the last three identities gives

$$\begin{aligned}
&\sum_{k=0}^{n} \binom{2n+1}{2k+1}\binom{2n+2k}{2k}2^{2n-2k}(-1)^k E_{2k+1} \\
&= 2^{2n}\sum_{k=0}^{2n} \binom{2n+1}{k+1}\binom{2n+k}{k}(-1)^k \\
&= 0,
\end{aligned}$$

where the last equality follows from Lemma 6.14. □

The third identity we present is similar in form to Sun's secant-number identity (1.6).

Theorem 6.15. *For $n \geqslant 0$, we have*

$$\sum_{k=0}^{n} \binom{2n}{2k} \binom{2n+2k+2}{2k+1} 2^{2n-2k-1} (-1)^k E_{2k} = (-1)^n \binom{2n}{n}. \tag{6.29}$$

The proof of Theorem 6.15 relies on the following different generalization of Theorem 6.1.

Theorem 6.16. *Fix an integer $d \geqslant 0$. Let $y_k = \sum_{l=0}^{k} \binom{k}{l} x_l$. Then,*

$$\sum_{k=0}^{n} \binom{n}{k} \binom{n+k+2d}{k+d} x_k = \sum_{k=0}^{n} \binom{n}{k} \binom{n+k+2d}{k+d} (-1)^{n-k} y_k. \tag{6.30}$$

Proof. If we set $x_k = \tilde{x}_k \prod_{i=0}^{d-1}(k-i)$ and $y_k = \tilde{y}_k \prod_{i=0}^{d-1}(k-i)$ in (6.4) and divide both sides by $\prod_{i=0}^{d-1}(n-i)$, then we obtain (6.30). □

Lemma 6.17. *For $n \geqslant 0$, we have*

$$\sum_{k=0}^{2n} \binom{2n}{k} \binom{2n+k+2}{k+1} (-1)^k 2^{2n-k-1} = (-1)^n \binom{2n}{n}. \tag{6.31}$$

Proof. This can be proved by using the same idea as in the proof of Lemma 6.9. The details are left to the interested reader. □

Inspired by Lemma 6.17, we find the following new alternating sum formula for the ubiquitous *Catalan numbers* $C_n = \frac{1}{n+1}\binom{2n}{n}$.

Proposition 6.18. *For $n \geqslant 0$, we have*

$$\sum_{k=0}^{2n} \binom{2n}{k} \binom{2n+k+2}{k+2} (-1)^k 2^{2n-k} = (-1)^n C_n. \tag{6.32}$$

Proof. This can again be proved by using the same idea as in the proof of Lemma 6.9. The details are left to the interested reader. □

We can now prove Theorem 6.15.

Proof of Theorem 6.15. By (6.30) with $d = 1$, we have

$$\sum_{k=0}^{2n} \binom{2n}{k} \binom{2n+k+2}{k+1} x_k = \sum_{k=0}^{2n} \binom{2n}{k} \binom{2n+k+2}{k+1} (-1)^k y_k, \tag{6.33}$$

where $y_k = \sum_{l=0}^{k} \binom{k}{l} x_l$. Setting $x_{2k} = 2^{2n-2k}(-1)^k E_{2k}$ and $x_{2k+1} = 0$ for $k \geqslant 0$ in (6.33) yields

$$\sum_{k=0}^{n} \binom{2n}{2k} \binom{2n+2k+2}{2k+1} 2^{2n-2k} (-1)^k E_{2k} = \sum_{k=0}^{2n} \binom{2n}{k} \binom{2n+k+2}{k+1} (-1)^k y_k, \tag{6.34}$$

where for $k \geqslant 0$,

$$y_{2k} = 2^{2n-2k+1} + (-1)^{k-1} 2^{2n-2k} E_{2k} \quad \text{by (3.13)}$$

and

$$y_{2k+1} = 2^{2n-2k} \quad \text{by (3.12)}.$$

Combining the last three equalities gives

$$\begin{aligned}
&\sum_{k=0}^{n}\binom{2n}{2k}\binom{2n+2k+2}{2k+1}2^{2n-2k}(-1)^k E_{2k}\\
&=\sum_{k=0}^{n}\binom{2n}{2k}\binom{2n+2k+2}{2k+1}2^{2n-2k}-\sum_{k=0}^{n-1}\binom{2n}{2k+1}\binom{2n+2k+3}{2k+2}2^{2n-2k-1}\\
&=\sum_{k=0}^{2n}\binom{2n}{k}\binom{2n+k+2}{k+1}(-1)^k 2^{2n-k}\\
&=2(-1)^n\binom{2n}{n},
\end{aligned}$$

where the last equality follows from Lemma 6.17. □

Lemma 6.19. *For $n\geqslant 1$, we have*

$$\sum_{k=0}^{n}\binom{2n}{2k}2^{2n-2k}(-1)^k E_{2k+1}=(-1)^{n+1}E_{2n+1} \tag{6.35}$$

and

$$\sum_{k=0}^{n}\binom{2n+1}{2k}2^{2n-2k}(-1)^k E_{2k+1}=(-1)^{n}E_{2n+1}. \tag{6.36}$$

Consequently,

$$\sum_{k=1}^{n}\binom{2n}{2k-1}2^{2n-2k-1}(-1)^k E_{2k+1}=(-1)^{n}E_{2n+1}. \tag{6.37}$$

Proof. By (3.11), we have

$$\begin{aligned}
2^{2n+1}&=\sum_{k=0}^{n}\binom{2n+1}{2k+1}2^{2n-2k}(-1)^k E_{2k+1}+(-1)^n E_{2n+1}\\
&=\sum_{k=0}^{n}\left(\binom{2n}{2k+1}+\binom{2n}{2k}\right)2^{2n-2k}(-1)^k E_{2k+1}+(-1)^n E_{2n+1}\\
&=2^{2n+1}+\sum_{k=0}^{n}\binom{2n}{2k}2^{2n-2k}(-1)^k E_{2k+1}+(-1)^n E_{2n+1},
\end{aligned}$$

which is equivalent to (6.35). Here we need to apply (3.10) to get the last equality.

Next, we proceed to deduce (6.36). By (3.10), we have

$$\begin{aligned}
2^{2n+1}&=\sum_{k=0}^{n}\binom{2n+2}{2k+1}2^{2n-2k}(-1)^k E_{2k+1}\\
&=\sum_{k=0}^{n}\left(\binom{2n+1}{2k+1}+\binom{2n+1}{2k}\right)2^{2n-2k}(-1)^k E_{2k+1}\\
&=2^{2n+1}-(-1)^n E_{2n+1}+\sum_{k=0}^{n}\binom{2n+1}{2k}2^{2n-2k}(-1)^k E_{2k+1},
\end{aligned}$$

which is equivalent to (6.36). Here we need to apply (3.11) to get the last equality.

Finally, identity (6.37) follows by subtracting (6.35) from (6.36). □

Remark 6.20. *Two different q-analogs of* (6.36) *were derived by Kim and Lin [21] from the combinatorial perspective.*

Theorem 6.21. *For $n \geqslant 1$, we have*

$$\sum_{k=0}^{n}\left(2\binom{2n}{2k}\binom{2n+2k}{2k}+\binom{2n}{2k+1}\binom{2n+2k+1}{2k+1}\right)2^{2n-2k}(-1)^k E_{2k+1}=2^{2n+1} \tag{6.38}$$

and

$$\sum_{k=0}^{n}\left(\binom{2n}{2k}\binom{2n+2k}{2k}+\binom{2n}{2k+1}\binom{2n+2k}{2k+1}\right)2^{2n-2k}(-1)^k E_{2k+1}=2^{2n+1}. \tag{6.39}$$

Proof. Setting $x_{2k}=2^{n-2k}(-1)^k E_{2k+1}$ and $x_{2k+1}=0$ for $k\geqslant 0$ in (6.4) yields

$$\sum_{k\geqslant 0}\binom{n}{2k}\binom{n+2k}{2k}2^{n-2k}(-1)^k E_{2k+1}=\sum_{k\geqslant 0}\binom{n}{k}\binom{n+k}{k}(-1)^{n-k}y_k, \tag{6.40}$$

where $y_0=2^n$,

$$y_{2k}=\sum_{l=0}^{k}\binom{2k}{2l}2^{n-2l}(-1)^l E_{2l+1}=(-1)^{k+1}2^{n-2k}E_{2k+1} \quad \text{by (6.35)} \tag{6.41}$$

for $k\geqslant 1$ and

$$y_{2k+1}=\sum_{l=0}^{k}\binom{2k+1}{2l}2^{n-2l}(-1)^l E_{2l+1}=(-1)^k 2^{n-2k}E_{2k+1} \quad \text{by (6.36)} \tag{6.42}$$

for $k\geqslant 0$. Combining (6.40), (6.41) and (6.42) gives

$$\begin{aligned}&\sum_{k\geqslant 0}\binom{n}{2k}\binom{n+2k}{2k}2^{n-2k}(-1)^k E_{2k+1}\\ &=(-1)^n 2^{n+1}+\sum_{k\geqslant 0}\binom{n}{2k}\binom{n+2k}{2k}(-1)^{n-k+1}2^{n-2k}E_{2k+1}\\ &\quad-\sum_{k\geqslant 0}\binom{n}{2k+1}\binom{n+2k+1}{2k+1}(-1)^{n-k}2^{n-2k}E_{2k+1},\end{aligned}$$

which is equivalent to

$$\begin{aligned}&(1+(-1)^n)\sum_{k\geqslant 0}\binom{n}{2k}\binom{n+2k}{2k}2^{n-2k}(-1)^k E_{2k+1}\\ &=(-1)^n 2^{n+1}-\sum_{k\geqslant 0}\binom{n}{2k+1}\binom{n+2k+1}{2k+1}2^{n-2k}(-1)^{n-k}E_{2k+1}.\end{aligned}$$

Substituting $n\leftarrow 2n$ in the above identity yields (6.38). Combining (6.38) with (6.26) produces (6.39). □

7. Further work

In this work, we provide involutive proofs of various combinatorial identities featuring Euler numbers. For future research, we mention the following three open problems.

(1) Our first combinatorial proof of Sun's secant number identity (1.6) consists of three steps:
   (a) An involutive proof of (6.3).
   (b) Combinatorial proofs of (3.12) and (3.13).
   (c) An involutive proof of (6.14).
   It would be interesting to find a direct involutive proof of (1.6).

(2) The problem of deriving q-analogs of Sun's secant number identity (1.6) and its three companion identities (6.24), (6.26) and (6.29) remains open. Even though a q-analog of (6.3) exists in the work of Guo and Zeng [16], we do not know how to use it to derive a q-analog of (1.6).

(3) Can Conjecture 5.13 be proven? In particular, it would be interesting to find a combinatorial interpretation for $B_{2n+1}(q)$.

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(Shishuo Fu) College of Mathematics and Statistics, Chongqing University & Key Laboratory of Nonlinear Analysis and its Applications (Chongqing University), Ministry of Education, Chongqing 401331, P.R. China
Email address: fsshuo@cqu.edu.cn

(Yang Li) Research Center for Mathematics and Interdisciplinary Sciences, Shandong University & Frontiers Science Center for Nonlinear Expectations, Ministry of Education, Qingdao 266237, P.R. China
Email address: 202421349@mail.sdu.edu.cn

(Zhicong Lin) Research Center for Mathematics and Interdisciplinary Sciences, Shandong University & Frontiers Science Center for Nonlinear Expectations, Ministry of Education, Qingdao 266237, P.R. China
Email address: linz@sdu.edu.cn

(Yongchun Zang) Research Center for Mathematics and Interdisciplinary Sciences, Shandong University & Frontiers Science Center for Nonlinear Expectations, Ministry of Education, Qingdao 266237, P.R. China
Email address: 202521437@mail.sdu.edu.cn